SI Appendix

Survival-guided matrix factorization identifies generalizable prognostic programs in pancreatic cancer

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Contents

List of Figures

List of Tables

Part I

Supplementary Methods

1 Model details

Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix with p genes and n subjects. DeSurv approximates $X \approx WH$ with $W \in \mathbb{R}_{\geq 0}^{p\times k}$ and $H \in \mathbb{R}_{\geq 0}^{k\times n}$, and links the shared program matrix W to survival via an elastic-net–penalized Cox model. Supervision is placed on the program matrix W, the basis that transfers to new cohorts, whereas the sample loadings H serve only to reconstruct the training expression matrix; new samples are therefore scored by projection onto the learned programs, $Z = W^\top X$, rather than by re-estimating loadings.

Let $y \in \mathbb{R}_{>0}^{n}$ be the observed survival times, $\delta \in \{0,1\}^n$ the event indicators, and $Z = W^\top X \in \mathbb{R}^{k\times n}$ the sample-wise factor scores. We write Z_i for the ith column of Z, and $n_{\text{event}} = \sum_{i=1}^n \delta_i$. For convenience we recall the DeSurv loss from Equation 1 in the main text and rewrite it as

$$\begin{aligned}\mathcal{L}(W,H,\beta) = \\ &(1-\alpha)\left(\frac{1}{2np}\|X - WH\|_F^2 + \frac{\lambda_H}{2nk}\|H\|_F^2\right) \\ &-\alpha\left(\frac{2}{n_{event}}\ell(W,\beta) - \lambda\{\xi\|\beta\|_1 + \tfrac{1-\xi}{2}\|\beta\|_2^2\}\right)\end{aligned}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\beta \in \mathbb{R}^k$ are the Cox coefficients. The Cox log-partial likelihood $\ell(W,\beta)$ is

$$\ell(W,\beta) = \sum_{i=1}^{n} \delta_i \left[Z_i^\top \beta - \log\left(\sum_{j:y_j \geq y_i} \exp(Z_j^\top \beta)\right)\right]. \tag{1}$$

The inner sum runs over the risk set $\{j : y_j \geq y_i\}$ of subjects still at risk at the observed time y_i; tied event times are handled by the Breslow approximation in the `glmnet` Coxnet routine used for the β update (below). Hyperparameters satisfy $\alpha \in [0,1)$, $\lambda_H \geq 0$, $\lambda \geq 0$, and $\xi \in [0,1]$. The constants $1/(2np)$, $2/n_{\text{event}}$, and $1/(2nk)$ place the reconstruction, loadings-penalty, and survival terms on comparable numerical scales for stable optimization.

2 Optimization Algorithm

2.1 Block coordinate descent scheme

Optimization proceeds by block coordinate descent (BCD; Algorithm S1), which at each outer iteration updates H, then W, then β while holding the other blocks fixed.

2.2 Update for H

Conditional on W, the loss $\mathcal{L}(W,H,\beta)$ reduces to a strictly convex quadratic function of H. We adopt the standard NMF multiplicative update with ℓ_2 penalty:

$$H \leftarrow \max\left(H \odot \frac{W^\top X}{W^\top WH + \lambda_H H + \varepsilon_H}, \varepsilon_H\right), \tag{2}$$

Algorithm S1 DeSurv block coordinate descent

Input: $X \in \mathbb{R}^{p\times n}_{\geq 0}$, survival times $y \in \mathbb{R}^n_{>0}$, event indicators $\delta \in \{0,1\}^n$, tolerance tol, maximum iterations $maxit$, supervision parameter α, rank k, penalties λ_H, λ, and ξ

Output: Fitted DeSurv parameters (W, H, β)

Initialize $W^{(0)}, H^{(0)}$ with positive entries (e.g., $W^{(0)}, H^{(0)} \sim \text{Uniform}(0, \max X)$)
Initialize $\beta^{(0)}$ (e.g., $\beta^{(0)} = \mathbf{1}$)
$loss = \mathcal{L}(W^{(0)}, H^{(0)}, \beta^{(0)})$
$eps = \infty$, $t = 0$
while $eps \geq tol$ **and** $t < maxit$ **do**
    (H-update)
        Update $H^{(t+1)}$ from $H^{(t)}$ using the multiplicative rule in Eq. 2,
        holding $W^{(t)}$ and $\beta^{(t)}$ fixed.
    (W-update)
        Update $W^{(t+1)}$ from $W^{(t)}$ using the hybrid multiplicative rule in Eq. 6,
        holding $H^{(t+1)}$ and $\beta^{(t)}$ fixed, and perform backtracking to ensure
        $\mathcal{L}$ does not increase.
    (β-update)
        Update $\beta^{(t+1)}$ from $\beta^{(t)}$ using a Newton-like step for the Cox loss
        (Eq. 7), holding $W^{(t+1)}$ and $H^{(t+1)}$ fixed.
    $lossNew = \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)})$
    $eps = |lossNew - loss|/|loss|$
    $loss = lossNew$
    $t = t + 1$
end while
return $\hat{W} = W^{(t+1)}, \hat{H} = H^{(t+1)}, \hat{\beta} = \beta^{(t+1)}$

where $\odot$ denotes elementwise multiplication, all divisions are elementwise, and $\varepsilon_H > 0$ is a small floor to prevent entries from becoming exactly zero. This update is equivalent to a majorization–minimization step and guarantees a nonincreasing reconstruction term conditional on W[1,2]. The ε_H-floor ensures that iterates remain in the interior of the nonnegative orthant, which simplifies the convergence analysis.

2.3 Update for W

For W, we construct a hybrid multiplicative update that balances the contributions of the NMF and Cox gradients. Let

$$\nabla_W \mathcal{L}_{\text{NMF}}(W, H) = \frac{1}{np}(WH - X)H^\top,$$

and let

$$\nabla_W \mathcal{L}_{\text{Cox}}(W, \beta) = \frac{2}{n_{event}} \nabla_W \ell(W, \beta).$$

where $\nabla_W \ell(W, \beta)$ denotes the Cox gradient with respect to W. A closed-form expression for $\nabla_W \ell$ is

$$\nabla_W \ell(W, \beta) = \sum_{i=1}^{n} \delta_i \left(x_i - \frac{\sum_{j: y_j \geq y_i} x_j \exp(x_j^\top W\beta)}{\sum_{j: y_j \geq y_i} \exp(x_j^\top W\beta)} \right) \beta^\top, \tag{3}$$

where x_i denotes the ith column of X.

We define the gradient-balancing factor

$$\zeta^{(t)} = \frac{\left\| (1-\alpha)\nabla_W \mathcal{L}_{\text{NMF}}(W^{(t)}, H^{(t+1)}) \right\|_F^2}{\left\| \alpha \nabla_W \mathcal{L}_{Cox}(W^{(t)}, \beta^{(t)}) \right\|_F^2}, \tag{4}$$

and clip $\zeta^{(t)}$ to avoid extreme values: $\zeta^{(t)} \leftarrow \min(\zeta^{(t)}, \zeta_{\max})$ with $\zeta_{\max} = 10^6$ in all experiments.

The multiplicative factor for W is then

$$R^{(t)} = \frac{\frac{(1-\alpha)}{np} X H^{(t+1)\top} + \frac{2\alpha}{n_{\text{event}}} \zeta^{(t)} \nabla_W \ell(W^{(t)}, \beta^{(t)})}{\frac{(1-\alpha)}{np} W^{(t)} H^{(t+1)} H^{(t+1)\top} + \varepsilon_W}, \tag{5}$$

and the proposed update is

$$W^{(t+1)} = \max\Big(W^{(t)} \odot R^{(t)}, \varepsilon_W\Big), \tag{6}$$

with a small floor $\varepsilon_W > 0$. When $\alpha = 0$, this reduces to the standard multiplicative update for NMF[1]; for $\alpha > 0$, the Cox gradient perturbs the update in a direction that decreases the supervised loss.

Because the combined multiplicative step does not, by itself, guarantee a nonincrease in the full loss $\mathcal{L}$, we embed it in a backtracking line search.

Algorithm S2 W update with backtracking

Input: Value of W and the previous iteration t: $W^{(t)}$, the multiplicative update term at the current iteration $R^{(t+1)}$, the max number of backtracking updates max_bt, the backtracking parameter $\rho \subset (0,1)$
Output: $W^{(t+1)}$

$\theta = 1$
b = 1
$flag\_accept = FALSE$
while $b \leq max\_bt$ **do**
    $W^{(t+1)}_{cand} = W^{(t)} \odot [(1-\theta) + \theta R^{(t+1)}]$
    (Column normalization)
        Compute $D = \mathrm{diag}\big(\|W^{(t+1)}_{(:,1)cand}\|_2, \ldots, \|W^{(t+1)}_{(:,k)cand}\|_2\big)$
        and set $W^{(t+1)}_{cand} \leftarrow W^{(t+1)}_{cand} D^{-1}$, $H^{(t+1)}_{cand} \leftarrow D H^{(t+1)}$, and $\beta^{(t)}_{cand} \leftarrow D\beta^{(t)}$
    if $\mathcal{L}(W^{(t+1)}_{cand}, H^{(t+1)}, \beta^{(t)}) \leq \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)})$ **then**
        $W^{(t+1)} = W^{(t+1)}_{cand}$
        $H^{(t+1)} = H^{(t+1)}_{cand}$
        $\beta^{(t)} = \beta^{(t)}_{cand}$
        $flag\_accept = TRUE$
        break
    end if
    $\theta = \theta * \rho$
    $b = b + 1$
end while
if $flag\_accept = FALSE$ **then**
    $W^{(t+1)} = W^{(t)}$
end if

The column normalization preserves both WH and $W\beta$, and therefore leaves the loss $\mathcal{L}$ invariant up to numerical error. Backtracking guarantees that the accepted W update does not increase $\mathcal{L}$.

2.4 Update for β

Conditional on (W, H), the loss in β reduces to a convex elastic-net–penalized Cox problem:

$$\min_{\beta \in \mathbb{R}^k} \left\{ -\frac{2\alpha}{n_{\text{event}}} \ell(W, \beta) + \lambda \left(\xi \|\beta\|_1 + \frac{1-\xi}{2} \|\beta\|_2^2 \right) \right\}.$$

We solve this subproblem by cyclic coordinate descent following[3]. Writing $\ell(\beta) = \ell(W, \beta)$ and $\tilde{\eta}_i = Z_i^\top \beta$, the update for coordinate r has the closed form

$$\hat{\beta}_r = \frac{S\left(\frac{1}{n}\sum_{i=1}^n w(\tilde{\eta})_i z_{i,r}\left[v(\tilde{\eta})_i - \sum_{j\neq r} z_{i,j}\beta_j\right], \lambda\xi\right)}{\frac{1}{n}\sum_{i=1}^n w(\tilde{\eta})_i z_{i,r}^2 + \lambda(1-\xi)}, \tag{7}$$

where $S(a, \tau) = \mathrm{sign}(a)\max(|a| - \tau, 0)$ is the soft-thresholding operator, and $w(\tilde{\eta})$, $v(\tilde{\eta})$ are standard local quadratic approximations of the Cox log-partial likelihood[3]. We iterate coordinate updates until the subproblem converges to numerical tolerance, yielding a minimizer in β for the current W.

2.5 Normalization of W

After each accepted W update, we normalize the columns of W as $W \leftarrow WD^{-1}$, and adjust H and β as

$$H \leftarrow DH, \qquad \beta \leftarrow D\beta,$$

where D is the diagonal matrix of column ℓ_2-norms of W. This preserves the reconstruction WH and the linear predictor $W\beta$, leaving both the NMF and Cox terms in the loss unchanged. Normalization prevents degeneracy in the scale-nonidentifiable factorization and keeps columns of W comparable in magnitude and interpretable as gene programs.

2.6 Derivation of W update from projected coordinate descent

The W update can be derived directly from the projected coordinate descent update with a specific step size.

Recall that the overall loss function is

$$\mathcal{L}(W, H, \beta) = \frac{(1-\alpha)}{2np}||X - WH||_F^2 - \frac{2\alpha}{n_{event}}\ell(W, \beta) + \lambda\left(\xi\|\beta\|_1 + \frac{(1-\xi)}{2}\|\beta\|_2^2\right).$$

where $\ell(W, \beta)$ is the log-partial likelihood for the Cox model. Let $\nabla_W \ell$ represent the derivative of $\ell(W, \beta)$ with respect to W. Then the derivative of the overall loss with respect to W is

$$\frac{\partial L}{\partial W} = \frac{(1-\alpha)}{np}\left(WHH^T - XH^T\right) - \frac{2\alpha}{n_{event}}\nabla_W \ell.$$

Then the gradient descent update rule at iteration t is

$$\begin{aligned} W^{(t)} &= W^{(t-1)} - \gamma\left(\frac{\partial L}{\partial W}\right) \\ &= W^{(t-1)} - \gamma\left(\frac{(1-\alpha)}{np}\left(WHH^T - XH^T\right) - \frac{2\alpha}{n_{event}}\nabla_W \ell\right). \end{aligned}$$

Let the step size γ be defined as in[1]:

$$\gamma = \frac{np}{(1-\alpha)}\frac{W}{WHH^T}.$$

Then the update becomes

$$\begin{aligned}
W^{(t)} &= W^{(t-1)} - \frac{np}{(1-\alpha)} \frac{W}{WHH^T} \left(\frac{(1-\alpha)}{np} \left(WHH^T - XH^T \right) - \frac{2\alpha}{n_{event}} \nabla_W \ell \right) \\
&= \frac{W}{WHH^T} XH^T + \frac{2\alpha np}{n_{event}(1-\alpha)} \nabla_W \ell \\
&= W \odot \frac{XH^T + \frac{2\alpha np}{n_{event}(1-\alpha)} \nabla_W \ell}{WHH^T} \\
&= W \odot \frac{\frac{(1-\alpha)}{np} XH^T + \frac{2\alpha}{n_{event}} \nabla_W \ell}{\frac{(1-\alpha)}{np} WHH^T}.
\end{aligned}$$

Finally, projected coordinate descent projects the W update in to the positive space

$$W^{(t)} = \max \left(W \odot \frac{\frac{(1-\alpha)}{np} XH^T + \frac{2\alpha}{n_{event}} \nabla_W \ell}{\frac{(1-\alpha)}{np} WHH^T}, 0 \right).$$

Since $W \geq 0$ this is equivalent to

$$W^{(t)} = W \odot \max \left(\frac{\frac{(1-\alpha)}{np} XH^T + \frac{2\alpha}{n_{event}} \nabla_W \ell}{\frac{(1-\alpha)}{np} WHH^T}, 0 \right),$$

which matches the multiplicative form used in the software implementation.

3 Convergence proof

We show that, under mild regularity conditions, the block coordinate descent (BCD) Algorithm S1 converges to a stationary point of the DeSurv loss function

$$\mathcal{L}(W, H, \beta).$$

For clarity, we first analyze the algorithm *without* the column normalization of W inside the loop, and then argue in Section 3.1 that the normalization step preserves stationarity of limit points.

Throughout, let $\theta = (W, H, \beta)$ and denote $\mathcal{L}(\theta) = \mathcal{L}(W, H, \beta)$. The feasible set is

$$\Theta = \{(W, H, \beta) : W \in \mathbb{R}_{\geq 0}^{p \times k},\ H \in \mathbb{R}_{\geq 0}^{k \times n},\ \beta \in \mathbb{R}^k\}.$$

Assumption 1 (Regularity and parameter space). We assume:

(i) The data $X \in \mathbb{R}_{\geq 0}^{p \times n}$, $y \in \mathbb{R}_{>0}^n$, and $\delta \in \{0, 1\}^n$ are fixed and bounded.

(ii) The hyperparameters satisfy $\lambda_H > 0$, $\lambda > 0$, $\alpha \in [0, 1)$, and $\xi \in [0, 1]$.

(iii) The initial iterate $\theta^{(0)} = (W^{(0)}, H^{(0)}, \beta^{(0)})$ lies in Θ and satisfies $W^{(0)}, H^{(0)} > 0$ elementwise.

Assumption 2 (Block updates). At each outer iteration t, the three block updates in Algorithm S1 satisfy:

(i) *H-update.* For fixed $(W^{(t)}, \beta^{(t)})$, the update $H^{(t)} \mapsto H^{(t+1)}$ is given by the multiplicative rule

$$H \leftarrow H \odot \frac{W^\top X}{W^\top WH + \lambda_H H},$$

applied at $(W^{(t)}, H^{(t)})$. This rule preserves nonnegativity and yields a nonincreasing value of the reconstruction term conditional on $W^{(t)}$ and $\beta^{(t)}$; see e.g. (Lee and Seung 2001; Pascual-Montano et al. 2006; Lin 2007).

(ii) W-*update.* For fixed $(H^{(t+1)}, \beta^{(t)})$, the update $W^{(t)} \mapsto W^{(t+1)}$ is the hybrid multiplicative step followed by backtracking as in Algorithm S2, producing $W^{(t+1)}$ such that

$$\mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t)}) \le \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)}).$$

If no candidate satisfies the Armijo-type condition within the allowed number of backtracking steps, the algorithm sets $W^{(t+1)} := W^{(t)}$.

(iii) β-*update.* For fixed $(W^{(t+1)}, H^{(t+1)})$, the update $\beta^{(t)} \mapsto \beta^{(t+1)}$ is obtained by running the coordinate descent method of Simon et al. (2011) on the convex elastic-net Cox subproblem until convergence. Thus

$$\beta^{(t+1)} \in \arg\min_{\beta \in \mathbb{R}^k} \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta),$$

and

$$\mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)}) \le \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t)}).$$

Lemma 1 (Continuity and bounded level sets). Under Assumption 1, $\mathcal{L} : \Theta \to \mathbb{R}$ is continuous, and the initial sublevel set

$$\mathcal{S}_0 := \{\theta \in \Theta : \mathcal{L}(\theta) \le \mathcal{L}(\theta^{(0)})\}$$

is nonempty, closed, and bounded.

Proof. The NMF term $\|X - WH\|_F^2$ is a polynomial in the entries of W and H, hence is continuously differentiable. The penalty $\|H\|_F^2$ is continuous as well. The Cox partial log-likelihood is a smooth function of the linear predictor $\eta_i = x_i^\top W\beta$, which is linear in (W, β), so this term is continuously differentiable in (W, β). The ℓ_2 penalty on β is smooth, and the ℓ_1 penalty is convex and lower semicontinuous. Therefore $\mathcal{L}$ is continuous on Θ.

The constraints $W, H \ge 0$ define closed convex cones, and $\beta \in \mathbb{R}^k$ is unconstrained. Because $\lambda_H > 0$ and $\lambda > 0$, the terms $\frac{\lambda_H}{nk}\|H\|_F^2$ and $\lambda\frac{1-\xi}{2}\|\beta\|_2^2$ dominate the objective as $\|H\|_F \to \infty$ or $\|\beta\|_2 \to \infty$, respectively. Thus the sublevel set $\mathcal{S}_0$ is bounded in (H, β). Since $W \ge 0$ and X is fixed, the reconstruction term and Cox term being bounded prevent W from diverging while $\mathcal{L}$ remains below $\mathcal{L}(\theta^{(0)})$, so W is bounded on $\mathcal{S}_0$. Because Θ is closed and $\mathcal{L}$ is continuous, $\mathcal{S}_0$ is closed. Nonemptiness follows from $\theta^{(0)} \in \mathcal{S}_0$. ☐

Lemma 2 (Monotone descent and existence of limit points). Under Assumptions 1 and 2, Algorithm S1 (without W normalization) generates a sequence $\{\theta^{(t)}\}_{t\ge 0} \subset \mathcal{S}_0$ such that

$$\mathcal{L}(\theta^{(t+1)}) \le \mathcal{L}(\theta^{(t)}) \quad \forall\, t,$$

and $\{\mathcal{L}(\theta^{(t)})\}$ converges to a finite limit $\mathcal{L}^*$. Moreover, $\{\theta^{(t)}\}$ is bounded and therefore admits at least one limit point.

Proof. By Assumption 2(i),

$$\mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)}) \le \mathcal{L}(W^{(t)}, H^{(t)}, \beta^{(t)}).$$

By Assumption 2(ii),

$$\mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t)}) \le \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)}).$$

By Assumption 2(iii),

$$\mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)}) \le \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t)}).$$

Combining these inequalities gives

$$\mathcal{L}(\theta^{(t+1)}) \le \mathcal{L}(\theta^{(t)}) \quad \forall t,$$

so $\{\mathcal{L}(\theta^{(t)})\}$ is monotonically nonincreasing. Since $\mathcal{L}$ is bounded below on Θ, the sequence converges to some finite $\mathcal{L}^*$.

Each iterate lies in $\mathcal{S}_0$ by construction, and $\mathcal{S}_0$ is bounded by Lemma 1. Therefore $\{\theta^{(t)}\}$ is bounded and has at least one limit point by the Bolzano–Weierstrass theorem. ☐

Lemma 3 (Blockwise optimality of limit points). Let $\theta^* = (W^*, H^*, \beta^*)$ be any limit point of $\{\theta^{(t)}\}$ under Assumptions 1–2. Then:

(i) H^* satisfies the KKT conditions for

$$\min_{H \geq 0} \mathcal{L}(W^*, H, \beta^*).$$

(ii) W^* satisfies the KKT conditions for

$$\min_{W \geq 0} \mathcal{L}(W, H^*, \beta^*).$$

(iii) β^* is the unique minimizer of

$$\min_{\beta \in \mathbb{R}^k} \mathcal{L}(W^*, H^*, \beta),$$

and satisfies the KKT conditions for the elastic-net Cox subproblem.

Proof. Let $\{t_j\}$ be a subsequence such that $\theta^{(t_j)} \to \theta^* = (W^*, H^*, \beta^*)$. By continuity of $\mathcal{L}$, $\mathcal{L}(\theta^{(t_j)}) \to \mathcal{L}^*$.

(i) Conditional on (W, β), the H-subproblem is

$$\min_{H \geq 0} \frac{(1-\alpha)}{np}(\|X - WH\|_F^2 + \frac{\lambda_H}{nk}||H||_F^2) + \text{const in } H,$$

a strictly convex quadratic program. The multiplicative update for H is known to be a descent method whose fixed points coincide with the KKT points of this constrained subproblem[1,4]. Since the full objective $\mathcal{L}(\theta^{(t)})$ converges, the per-iteration decrease in $\mathcal{L}$ due to the H-update must vanish along $\{t_j\}$. If H^* were not KKT for the H-subproblem given (W^*, β^*), there would exist a feasible descent direction for H at (W^*, β^*), implying that for sufficiently large j the H-update would still produce a strict decrease in $\mathcal{L}$, contradicting convergence. Hence H^* satisfies the KKT conditions.

(ii) Conditional on (H, β), the W-subproblem is differentiable and convex in W (sum of a quadratic term and a negative Cox partial log-likelihood in a linear predictor). The hybrid multiplicative step with backtracking can be viewed as a projected gradient-like update onto the nonnegative orthant: as shown in the derivation above, the multiplicative direction is equivalent to a negative gradient step with the adaptive step size of Seung et al., projected onto the nonnegative orthant. Under mild conditions on the search direction and step sizes, any limit point of such a projected gradient scheme is a KKT point for the constrained subproblem; see, for example,[5,6]. If W^* did not satisfy the KKT conditions, there would be a feasible descent direction at (W^*, H^*, β^*) and a sufficiently small step that reduces $\mathcal{L}$, which the line search would eventually accept. This would contradict the fact that $\mathcal{L}(\theta^{(t_j)}) \to \mathcal{L}^*$. Therefore W^* is KKT for the W-subproblem.

(iii) For fixed (W, H), the β-subproblem is the convex elastic-net penalized Cox objective. The coordinate descent algorithm converges to the unique minimizer. By Assumption 2(iii),$\beta^{(t+1)}$ is taken to be this minimizer at each iteration. Passing to the limit along $\{t_j\}$ shows that β^* is the unique minimizer of the β-block given (W^*, H^*) and hence satisfies its KKT conditions. □

Theorem 1 (Convergence to a stationary point). Under Assumptions 1 and 2, the sequence $\{\theta^{(t)}\}$ produced by Algorithm S1 (without W normalization) satisfies:

(a) The objective values $\{\mathcal{L}(\theta^{(t)})\}$ are monotonically nonincreasing and converge to a finite limit $\mathcal{L}^*$.

(b) Every limit point $\theta^* = (W^*, H^*, \beta^*)$ of $\{\theta^{(t)}\}$ is a stationary point of $\mathcal{L}$ on Θ, i.e. it satisfies the first-order KKT conditions for

$$\min_{\theta \in \Theta} \mathcal{L}(\theta).$$

Proof. Part (a) is Lemma 2. For part (b), Lemma 3 shows that any limit point θ^* is blockwise optimal: each block is optimal (in the KKT sense) when the others are held fixed.

The DeSurv loss can be written as

$$\mathcal{L}(\theta) = f(W, H, \beta) + g(\beta) + I_{\{W \geq 0\}}(W) + I_{\{H \geq 0\}}(H),$$

where f is continuously differentiable, $g(\beta) = \lambda\xi\|\beta\|_1$ is a separable convex (possibly nondifferentiable) penalty, and I_C denotes the indicator of a closed convex set C. This is the smooth+nonsmooth composite form considered in block coordinate descent analyses such as[5]. In this setting, any point that is optimal with respect to each block individually (a block coordinatewise minimizer) is a stationary point for the full problem: equivalently,

$$0 \in \nabla_{W,H} f(W^*, H^*, \beta^*) + \partial I_{\{W \geq 0\}}(W^*) + \partial I_{\{H \geq 0\}}(H^*),$$

$$0 \in \nabla_\beta f(W^*, H^*, \beta^*) + \partial g(\beta^*),$$

which are precisely the KKT conditions for $\min_{\theta \in \Theta} \mathcal{L}(\theta)$. Hence every limit point of Algorithm S1 is stationary. □

3.1 Effect of column normalization of W

The above analysis omits the column-normalization step for W used in Algorithm S2. We briefly argue that this normalization does not affect stationarity of limit points.

Let D be a diagonal matrix with strictly positive diagonal entries. The transformation

$$(W, H, \beta) \mapsto (W', H', \beta') = (WD^{-1},\ DH,\ D\beta)$$

preserves both the product WH and the linear predictor $W\beta$, and hence leaves $\mathcal{L}$ invariant. The column-normalization step is exactly such a transformation, with D chosen from the column norms of W.

Let $\{\theta^{(t)}\}$ be the sequence generated by the algorithm without normalization, and let $\{\tilde{\theta}^{(t)}\}$ be the sequence with normalization. For each t there exists a diagonal $D^{(t)}$ with strictly positive entries such that

$$\tilde{\theta}^{(t)} = (W^{(t)} D^{(t),-1},\ D^{(t)} H^{(t)},\ D^{(t)} \beta^{(t)}),$$

and $\mathcal{L}(\tilde{\theta}^{(t)}) = \mathcal{L}(\theta^{(t)})$. Thus limit points of $\{\tilde{\theta}^{(t)}\}$ are obtained from limit points of $\{\theta^{(t)}\}$ by such invertible diagonal scalings.

Since the KKT conditions are expressed in terms of the gradients with respect to WH and $W\beta$, and these quantities are invariant under the above scaling, stationarity is preserved under the transformation. Therefore Theorem 1 implies that every limit point of the *normalized* algorithm is also a stationary point of $\mathcal{L}$ (up to this scaling equivalence), which establishes convergence of the implementation used in practice.

3.2 Remark (Coxnet implementation)

Our implementation updates the β block using a Coxnet-style coordinate descent that relies on an approximate Hessian. Consequently, the formal optimality proof is established for an idealized version of the β update that assumes exact second-order information. Nonetheless, in empirical applications the approximate update is numerically stable and yields a monotone decrease in the objective, consistent with the behavior predicted by the idealized analysis.

4 Supervision on W versus H

Classical nonnegative matrix factorization (NMF) is non-identifiable: for any invertible matrix R with nonnegative entries, the factorization (W, H) can be transformed to $(WR, R^{-1}H)$ without changing the reconstruction error, yielding multiple valid solutions[7–9]. Although normalizing columns of W or rows of

H resolves the trivial scaling ambiguity, nonnegative mixing transformations persist, and empirical studies consistently show that the sample-loading matrix H is more sensitive to initialization and local minima than the program matrix W[10–12].

In DeSurv, survival supervision enters through the projection $Z = W^\top X$ in the partial log-likelihood, so that the gradient of the DeSurv loss with respect to the programs is

$$\nabla \mathcal{L} = (1 - \alpha)\nabla_W \mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha \nabla_W \mathcal{L}_{\mathrm{Cox}}(W, \beta),$$

and therefore the update rule for W as in Equation 6 relies on both the reconstruction term and the supervision term. As such, the gene-program definitions are directly shaped by their association with survival. This mechanism amplifies genes aligned with the hazard gradient and suppresses those that dilute prognostic structure, ensuring that the learned programs encode survival-relevant biological variation.

By contrast, if supervision were applied to the sample loadings H, the model could reduce the Cox loss by redistributing patient-specific coefficients while leaving the gene-level programs W nearly unchanged, a behavior documented in multiple supervised variants of NMF and supervised topic models, where supervision applied only to the coefficient matrix modifies H but not the basis W[13,14].

Supervising W also provides a practical advantage for **portability**: because W defines gene-level programs, new samples can be embedded by the closed-form projection $Z_{\mathrm{new}} = X_{\mathrm{new}}^\top W$. In contrast, supervising H would not yield such a mapping; instead, obtaining sample loadings for external datasets would require solving a nonnegative least-squares (NNLS) problem for each new sample, introducing optimization noise and reducing reproducibility.

Together, these considerations motivate supervising through W, which shapes the gene programs toward prognostic directions, stabilizes solutions across initializations, and yields biologically interpretable signatures that generalize across cohorts.

5 Cross-validation procedure

Algorithm S3 describes the cross validation procedure with F folds and R initializations. We define the c-index using comparable pairs $\mathcal{P} = \{(i, j) : y_i < y_j,\ \delta_i = 1\}$ and linear predictors $\hat{\eta}_i$:

$$\hat{c} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left[\mathbb{1}(\hat{\eta}_i > \hat{\eta}_j) + \tfrac{1}{2}\mathbb{1}(\hat{\eta}_i = \hat{\eta}_j)\right]. \quad (8)$$

Algorithm S3 Cross-validation for DeSurv pipeline

Input:
Number of folds F
Number of initializations R
Observed data (X, y, δ)
Hyperparameters k, α, λ, ξ, and λ_H

Output: The cross-validated c-index

Divide subjects into F folds.
for $f = 1$ to F **do**
  Split data into training and validation $(X_{(-f)}, y_{(-f)}, \delta_{(-f)})$, and $(X_{(f)}, y_{(f)}, \delta_{(f)})$ sets
  for $r = 1$ to R **do**
    set.seed(r)
    Apply Algorithm S1 with inputs $(X_{(-f)}, y_{(-f)}, \delta_{(-f)})$, and $(k, \alpha, \lambda, \xi, \lambda_H)$
    Obtain $\hat{W}$ and $\hat{\beta}$ as output from Algorithm S1
    Compute the estimated linear predictor: $\hat{\eta} = X_{(f)}^T \hat{W} \hat{\beta}$
    Compute c-index, denoted $\hat{c}_{(f)r}$, according to Equation 8
  end for
  end loop over r
  Compute average c-index across initializations: $\hat{c}_{(f)} = \frac{1}{R} \sum_{r=1}^{R} \hat{c}_{(f)r}$
end for
end loop over f
Compute final cross-validated c-index: $\hat{c} = \frac{1}{F} \sum_{f=1}^{F} \hat{c}_{(f)}$
return Final estimate $\hat{c}$

6 Bayesian Optimization

We treat hyperparameter tuning as a black-box optimization problem over a bounded domain. Let θ collect the tuned parameters (at minimum $\theta = (k, \alpha, \lambda, \xi)$, and optionally discrete choices such as the signature size n_{top}). For a given θ we run the cross-validation procedure in Algorithm S3, averaging the c-index across folds and random initializations to form the objective

$$\hat{c}(\theta) = \frac{1}{F} \sum_{f=1}^{F} \hat{c}_{(f)}(\theta).$$

Because $\hat{c}(\theta)$ is expensive to evaluate and non-differentiable with respect to θ, we use Bayesian optimization (BO) to adaptively select candidate configurations.

We place a Gaussian-process (GP) prior on the response surface $f(\theta) \sim \mathcal{GP}(0, k(\theta, \theta'))$ with a Mat'ern kernel and automatic relevance determination length-scales. Continuous parameters are rescaled to $[0, 1]$ and penalties are optimized on a log scale; integer-valued coordinates (e.g., k or n_{top}) are proposed in the continuous space and rounded inside the evaluator. After an initial batch of evaluations, the GP is refit after each new observation and used to guide acquisition.

At iteration t, let $\mu_t(\theta)$ and $\sigma_t(\theta)$ denote the GP posterior mean and standard deviation, and let $f_\star$ be the best observed score. We select new candidates by maximizing expected improvement (EI),

$$\text{EI}(\theta) = (\mu_t(\theta) - f_\star - \kappa)\, \Phi(Z(\theta)) + \sigma_t(\theta) \phi(Z(\theta)), \quad Z(\theta) = \frac{\mu_t(\theta) - f_\star - \kappa}{\sigma_t(\theta)},$$

where Φ and ϕ are the standard normal cdf/pdf and $\kappa \geq 0$ controls the exploration margin. EI is evaluated over a candidate pool sampled from the bounded search region, and the maximizer is evaluated next. The loop continues until the iteration budget is exhausted or the improvement plateaus.

For model selection, we take the highest observed $\hat{c}(\theta)$ as the optimal configuration. When fold-level standard errors are available, we apply a one-standard-error rule for k, selecting the smallest rank whose mean performance lies within one standard error of the best. The final DeSurv model is refit at the selected parameters using the consensus-based initialization described above, and the BO-chosen n_{top} (when tuned) is used to truncate each program for downstream analysis.

The parameter n_{top} controls which genes enter the survival linear predictor. After factorization, each sample's score for factor j is computed as $\widetilde{W}_j^\top x$, where $\widetilde{W}_j$ retains only the n_{top} genes with the largest factor-specificity scores s_{ij} (SI Appendix, Section 9); genes outside this set still shape the factorization but do not enter the linear predictor. BO selects n_{top} jointly with k, α, λ, and ξ by optimizing cross-validated C-index. When $\alpha = 0$ (no survival supervision), gene selection and survival outcomes are decoupled, so varying n_{top} has negligible effect on precision in the null simulation.

7 PDAC Datasets

We analyzed PDAC cohorts from TCGA-PAAD and CPTAC-PDAC as training data[15,16]. External validation used independent cohorts from Dijk, Moffitt, PACA-AU (array and RNA-seq), and Puleo[17–20]. Training models were fit on the combined TCGA and CPTAC expression matrices after harmonizing gene identifiers; validation datasets were processed with the same gene set and transformations before prediction and clustering.

Dataset-specific inclusion and preprocessing steps were as follows. For TCGA-PAAD, we excluded samples lacking tumor grade and mapped features to gene symbols. For CPTAC, we retained samples annotated as PDAC by histology. For Dijk, all 90 samples passed quality-control filters and were retained. For Moffitt, only primary tumor specimens were kept. For PACA-AU array and RNA-seq, we retained primary PDAC tumors and collapsed duplicated gene symbols by median; the RNA-seq cohort received a "_seq' ' suffix on sample IDs to avoid collisions with the array cohort. Puleo required no additional dataset-specific filtering beyond survival QC.

Across cohorts, samples without valid survival outcomes (nonpositive time, missing time, or missing event indicator) were removed. Expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort (TCGA-PAAD and CPTAC) separately, genes were ranked by mean expression and by variance independently; the 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection of these two gene sets yielded 1,970 genes used for model training. Validation datasets were restricted to this same 1,970-gene set. Within-sample rank transformation was applied to all datasets to mitigate platform effects, with validation data transformed identically to training.

Table S1 summarizes the cohorts, platforms, sample sizes after quality control, and patient-level metadata used in this study. Overall survival time and event indicators were available for all cohorts. PurIST[21] and DeCAF[22] molecular classifications, applied to bulk expression data as described in those references, were used as covariates in the covariate-adjusted external validation analysis. Gene expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study[22]; training data were obtained directly from GDC.

8 Survival Analysis

Survival analyses used overall survival time with event indicators as provided by each cohort. For fitted DeSurv models, sample-level program scores were computed as $Z = \widetilde{W}^\top X$ and standardized using the training mean and standard deviation before constructing the linear predictor $\hat{\eta} = Z\hat{\beta}$. Prognostic performance was summarized using the concordance index as defined above. For visualization and group-level testing, we generated Kaplan–Meier curves and log-rank tests for risk groups defined by median splits of $\hat{\eta}$ within each dataset to avoid cross-cohort scaling differences. When multiple datasets were pooled, Cox models were stratified by dataset to allow cohort-specific baseline hazards. All p-values reported for survival differences between groups correspond to two-sided log-rank tests.

9 Gene Program Correlation Analysis

To assess the biological identity of learned factors, we evaluated the correspondence between factor-specific gene rankings and established PDAC gene programs. For each factor column j of the gene program matrix W, we identified the top 50 genes by factor specificity score. Each column of W was first scaled by its column maximum to produce $W^*_{ij} = W_{ij} / \max_i W_{ij}$, placing all factors on a common $[0, 1]$ scale. The specificity score for gene i in factor j was then defined as

$$s_{ij} = W^*_{ij} - \max_{j' \neq j} W^*_{ij'}.$$

Genes were ranked in descending order of s_{ij}, and the top 50 per factor were retained. The union of these gene sets across all factors was then intersected with the genes present in any reference program to form the set of genes used in the correlation analysis.

For each factor j and each reference gene program G_k, we computed the Spearman rank correlation between the continuous factor loadings $W_{\cdot j}$ (restricted to the intersection gene set) and a binary indicator vector $\mathbf{1}[\text{gene} \in G_k]$. P-values were derived from the asymptotic approximation to the Spearman test statistic and adjusted for multiple comparisons using the Benjamini–Hochberg procedure applied within each factor column independently. Gene programs were included in the visualization only if at least one factor yielded a Spearman correlation exceeding 0.2. Asterisks ($*$) indicate adjusted p-value < 0.1.

Reference gene programs included Classical and Basal-like tumor cell signatures from Moffitt et al. [18]; iCAF and myCAF cancer-associated fibroblast signatures from Elyada et al. [23]; promoting (proCAF) and restraining (restCAF) cancer-associated fibroblast subtypes from the DeCAF study[22]; proCAF and restCAF subtypes from SCISSORS[24]; tumor, stromal, and immune compartment signatures from DECODER[25]; and Classical and Basal-like subtype classifiers from PurIST[21]. Programs from Bailey et al. [19], periductal subtypes, and MSI subtypes were excluded from visualization due to redundancy with other included programs or low overall correlation across all factors.

10 Reconstruction and Survival Contribution per Factor

NMF does not provide an orthogonal variance decomposition; it seeks a low-rank reconstruction of the expression matrix $X \approx WH$. We therefore measure each factor's contribution to reconstruction rather than to "variance explained.'' For a subset of factors $S \subseteq \{1, \ldots, k\}$, define the reconstruction value

$$v(S) = \|X\|_F^2 - \left\| X - \sum_{j \in S} W_j H_j^\top \right\|_F^2,$$

the reduction in residual error relative to the null (zero-factor) model, where $W_j H_j^\top$ is the rank-one outer-product contribution of factor j. The reconstruction Shapley value of factor j,

$$\phi_j = \sum_{S \subseteq \{1,\ldots,k\} \setminus \{j\}} \frac{|S|!\,(k - |S| - 1)!}{k!} \big(v(S \cup \{j\}) - v(S)\big),$$

is its average marginal reconstruction gain over all orderings of the factors. By the Shapley efficiency axiom, $\sum_j \phi_j = v(\{1, \ldots, k\})$, the total reconstruction gain of the full model, so the normalized shares $\phi_j / \sum_l \phi_l$ partition the model's reconstruction and sum to exactly 1. This is a genuine additive decomposition, in contrast to per-factor variance fractions, which for non-orthogonal factorizations such as NMF do not partition (cross-terms between rank-one contributions are non-zero). The Shapley shares are computed exactly by enumerating the 2^k subsets (trivial at $k = 3$ or $k = 5$).

The survival contribution of factor j was quantified as the Type III partial likelihood increment: the reduction in Cox partial log-likelihood when factor j is omitted from the full k-factor model,

$$\Delta\ell_j = \ell(\beta_1, \ldots, \beta_k) - \ell(\beta_1, \ldots, \beta_{j-1}, \beta_{j+1}, \ldots, \beta_k),$$

where $\ell(\cdot)$ denotes the Cox partial log-likelihood evaluated at the maximum partial likelihood estimates. Larger $\Delta\ell_j$ indicates a greater marginal contribution of factor j to survival prediction given all other factors. These two quantities are plotted against each other in main text, Fig. 2C. In PDAC at $k = 3$, the reconstruction shares are 41.8%/18.6%/39.7% for NMF (N1/N2/N3) and 18.8%/25.2%/56.0% for DeSurv (D1/D2/D3), while the survival contribution $\Delta\ell$ concentrates in DeSurv D1 ($\Delta\ell = 66.4$) with all NMF factors negligible — the factors that contribute most to reconstruction are not the most prognostic.

11 Factor Correspondence Analysis

To assess which biological programs are preserved, altered, or suppressed under survival supervision, we quantified the pairwise correspondence between NMF and DeSurv factor loadings. For each pair of factors $(j_{\text{NMF}}, j_{\text{DeSurv}})$, we computed the Spearman rank correlation between the corresponding columns of the NMF gene program matrix W_{NMF} and the DeSurv gene program matrix W_{DeSurv} over all genes. We use the full loading profile rather than a top-gene subset, which would impose an arbitrary threshold and overstate apparent preservation. Spearman correlation was used to capture monotone correspondence in gene loading ranks while remaining robust to scale differences between the two methods. Results are displayed as a heatmap with NMF factors as rows and DeSurv factors as columns (main text, Fig. 2D).

11.1 Cross-method projection R^2: novelty versus inheritance

The correlation heatmap summarizes pairwise similarity; to make a direction-aware statement of how much of each factor is reconstructable from the *other* method's basis, we regressed each factor's gene-loading vector onto the full set of the other method's loading vectors and report the ordinary least-squares R^2 (Table S2). A high R^2 means the factor is inherited (reconstructable from the other basis); a low R^2 means it is novel.

The headline is DeSurv D1: $R^2 = 0.09$ from the NMF basis, i.e. essentially unreconstructable from any combination of NMF factors. The prognostic program DeSurv creates is therefore novel, not a reweighting of an existing unsupervised factor. Conversely, the biologically dominant tumor (N1) and stromal (N3) programs carry over intact ($R^2 = 0.90$ and 0.88), so supervision conserves nearly all of the unsupervised structure and spends its remaining capacity resolving tumor into Basal- and Classical-aligned programs and shedding the prognostically uninformative exocrine factor.

11.2 Overall reconstruction R^2: the price of supervision

DeSurv gives up only about 1.5 percentage points of reconstruction R^2 relative to unsupervised NMF (Table S3) — very little reconstruction fidelity is sacrificed in exchange for the prognostic reorganization documented above.

11.3 The D1 axis is recovered by survival supervision, not by DeSurv specifically

If the D1 tumour–stroma axis were an artifact of DeSurv's particular optimizer, independent survival-supervised methods should not recover it. We therefore fit three other supervised dimension-reduction methods on the same training cohort ($n = 273$, 139 events) — supervised principal components (Cox-screened genes followed by PCA), one-component Cox partial least squares against martingale residuals, and a sparse Cox model (lasso-penalized, 32 selected genes) — and compared their leading survival-aligned direction to the DeSurv D1 score. All three recovered D1 closely (per-patient $|r| = 0.81$ for supervised PCA, 0.83 for Cox-PLS, and 0.85 for sparse Cox; mean 0.83), and each aligned specifically with D1 rather than the stromal D2 or Basal-like D3 programs (Fig. S1). By contrast, the leading *unsupervised* direction (the first principal component, which dominant-variance methods such as standard NMF and DECODER follow) was nearly orthogonal to D1 ($|r| = 0.10$) and instead tracked the high-variance D2/D3 stroma–Basal axis. This D1 axis is thus a property of survival supervision shared across methods — consistent with its being unreconstructable from the unsupervised NMF basis (Table S2) and with standard NMF's failure to enrich for the proCAF stroma at matched rank — and not an idiosyncrasy of the DeSurv algorithm.

Proper-scoring sensitivity of the external validation

The main-text external validation reports discrimination with Harrell's C-index under a Cox proportional-hazards model. Because Harrell's C can be optimistic when hazards are non-proportional[26,27], we additionally evaluated the frozen DeSurv linear predictor with a proper score, the integrated Brier score (IBS), in each validation cohort, recalibrating only the scalar slope within each cohort and scoring over that cohort's follow-up. For a single time-invariant risk score under proportional hazards the monotone score-to-survival map preserves the concordance ordering, so Antolini's time-dependent concordance coincides with Harrell's C; the additive quantity is therefore the proper score, not a second concordance. Across the 5 cohorts the DeSurv predictor achieved an event-weighted integrated Brier score of 0.158 versus 0.167 for a Kaplan–Meier reference, an index of prediction accuracy ($\mathrm{IPA} = 1 - \mathrm{IBS}/\mathrm{IBS}_{\mathrm{null}}$) of 5.4% (per-cohort range 3.4–9.7%). This modest proper-score improvement over the null is consistent with the moderate discrimination ($C = 0.62$) and with DeSurv's contribution being source-attributable decomposition rather than higher predictive accuracy.

Independence of discovery and validation

To make the separation between discovery, adjustment, and validation unambiguous: DeSurv is trained with survival supervision only; it never uses PurIST, DeCAF, or any molecular subtype label during factorization or hyperparameter selection. The PurIST and DeCAF subtype calls were fixed before the DeSurv analysis (loaded as precomputed per-cohort annotations) and enter only as covariates in the adjusted Cox models, not as inputs to DeSurv. All rank and hyperparameter selection (including the one-standard-error rule) was performed by nested cross-validation entirely within the TCGA and CPTAC training data; no external validation cohort contributed to training, scoring, cutpoint selection, or tuning. Validation-cohort survival outcomes were used only for the final, out-of-sample performance evaluation.

Part II

Supplementary Results

12 The DeSurv model shows consistent convergence and stability across restarts

Across datasets and initialization schemes, the DeSurv algorithm consistently converged to numerically stable solutions within the designated iteration budget. Warm-start strategies, in which solutions at $\alpha = 0$ initialized supervised runs, substantially reduced variability across restarts and improved reproducibility. Fig. S2 shows representative loss trajectories demonstrating monotone decreases until convergence across restarts. Compared with naive random initialization, warm-starts produced tighter distributions of cross-validated C-index and partial likelihood, confirming stability of the optimization procedure.

13 Simulation results under null and mixed scenarios

To characterize the operating characteristics of DeSurv across different signal regimes, we evaluated performance under three simulation scenarios. In the prognostic scenario (main text Fig. 5), prognostic programs explained low variance relative to outcome-neutral background signals, and survival depended on 150 factor-specific marker genes ($\beta_1 = 2$, $\beta_2 = \beta_3 = 0$). In the null scenario, survival times were generated independently of gene expression ($\beta = 0$), so no prognostic structure existed. In the mixed scenario, survival depended on 300 genes per lethal factor, of which 50% were factor-specific markers and 50% were background genes with loadings shared across all factors. All scenarios used $G = 3{,}000$ genes, $N = 200$ samples, $K = 3$ true factors, and identical gamma distribution parameters for the gene loading matrix W (see Materials and Methods). Performance was evaluated using cross-validated concordance index (C-index), precision of recovered prognostic gene sets, and the distribution of selected ranks across simulation replicates.

Under the null scenario (Fig. S3A–B), DeSurv showed no advantage over unsupervised NMF ($\alpha = 0$). Both methods yielded C-index values near 0.5, consistent with the absence of survival signal. The distribution of selected ranks was concentrated at low values for both methods, with NMF more strongly favoring $k = 2$. These results confirm that DeSurv does not hallucinate prognostic structure when none exists: Bayesian optimization selected low values of the supervision parameter α, effectively recovering standard NMF as a special case. This specificity control is important because it establishes that the advantages observed in the prognostic scenario reflect genuine signal recovery rather than overfitting to noise in the survival labels.

Under the mixed scenario (Fig. S3C–F), four mechanistic observations characterize DeSurv's advantage. First, when precision is evaluated against the factor-specific marker genes alone (150 genes; Fig. S3D left), DeSurv substantially outperforms NMF—matching or exceeding prognostic-scenario levels—even though overall survival-gene precision (300-gene survival set including background genes; Fig. S3D right) is more modestly improved. This dissociation reveals that background-gene recovery is similar between methods, and DeSurv's edge comes specifically from concentrating marker programs. Second, the learned factor most aligned with the true marker program receives a substantially larger Cox coefficient ($|\beta|$) under DeSurv than under NMF (Fig. S3E), confirming that survival supervision preferentially amplifies the marker-specific signal. Third, C-index values are modestly higher for DeSurv (Fig. S3C), consistent with the moderate overall improvement when background-gene dilution attenuates the survival signal. Together, these results demonstrate that DeSurv's advantage in the mixed scenario is mechanistic: the method specifically recovers marker programs and assigns them elevated survival weights, rather than incidental capture of background-gene prognosis.

Together, the three scenarios provide a dose-response relationship: DeSurv's advantage is greatest when variance and prognosis diverge (prognostic), moderate when they partially overlap (mixed), and absent when no survival signal exists (null). The mechanistic basis for the attenuation in the mixed scenario is now clear: DeSurv continues to specifically recover marker programs (high marker precision, elevated $|\beta|$), but the overall survival-gene precision improvement is dampened because background-gene dilution in the 300-gene survival signal reduces the discrimination advantage. This gradient is consistent with the predictions of sufficient dimension reduction theory and offers practical guidance for users: DeSurv is most beneficial in settings where the dominant transcriptional axes are expected to be prognostically neutral, as is common in cancers with high tumor purity variation or dominant tissue-composition signals.

14 Cross-validated C-index across factorization rank

To evaluate how the number of latent factors k affects prognostic performance, we performed an exhaustive grid search over $k \in \{2, 3, \ldots, 12\}$ and supervision strength $\alpha \in \{0, 0.05, \ldots, 0.95\}$ using 5-fold cross-validation on the TCGA+CPTAC training cohort. For each k, the best-performing α was selected by maximizing the mean CV C-index. The unsupervised NMF baseline ($\alpha = 0$) was evaluated at the same k values for comparison. Results are shown for two gene-selection settings: $n_{\text{top}} = 270$ top genes per factor and $n_{\text{top}} = \text{ALL}$ (all genes retained), both at $\lambda = 0.349$ and $\xi = 0.056$.

Fig. S4 shows the resulting C-index curves for both training (CV) and external validation (pooled across held-out cohorts), with shaded ribbons indicating ± 1 standard error. In training, DeSurv consistently outperforms NMF across all k, confirming that survival supervision improves in-sample concordance. In external validation, DeSurv at $k = 3$ achieves concordance comparable to NMF at higher ranks (e.g., $k = 7$; Table S4), demonstrating that survival supervision concentrates prognostic information into fewer factors rather than requiring additional rank to capture it.

15 Standard NMF rank selection heuristics

Standard unsupervised rank selection criteria (reconstruction residuals, cophenetic correlation coefficient, and mean silhouette width) yielded inconsistent guidance for selecting the factorization rank k in the PDAC training cohort (TCGA + CPTAC). Reconstruction residuals decreased smoothly without a clear elbow (Fig. S5A). Cophenetic correlation fluctuated without a distinct transition point (Fig. S5B). Mean silhouette width favored very low ranks that conflicted with other criteria (Fig. S5C). These contradictions motivated the survival-supervised rank selection approach described in the main text.

16 Cutpoint selection and validation of dichotomized risk groups

To convert the continuous DeSurv linear predictor into a clinically interpretable binary risk stratification, we selected an optimal z-score cutpoint via cross-validation on the TCGA+CPTAC training cohort. For each candidate cutpoint in $\{-2.0, -1.8, \ldots, 2.0\}$, we computed the mean absolute log-rank z-statistic across held-out folds and selected the cutpoint maximizing this metric (Fig. S6A). We then applied this cutpoint to external validation cohorts by standardizing each patient's linear predictor using the training mean and standard deviation. Fig. S6B shows the resulting Kaplan–Meier curves for the pooled external validation analysis, with a stratified log-rank test accounting for dataset of origin. Panels C–F display per-cohort validation KM curves for Dijk, Moffitt, PACA-AU (array and RNA-seq subsets pooled), and Puleo, respectively. Across all cohorts, the DeSurv-derived high-risk group showed consistently shorter survival, confirming that the training-derived cutpoint generalizes to independent datasets.

17 Overlap of DeSurv risk groups with known molecular subtypes

To assess whether the DeSurv-derived risk stratification captures biologically meaningful transcriptional programs, we examined the overlap between dichotomized risk groups (High vs Low) and two established

PDAC molecular classifiers: PurIST (Basal-like vs Classical) and DeCAF (proCAF vs restCAF). Fig. S7 shows the composition of each risk group across all pooled external validation cohorts, with Fisher's exact test P-values quantifying the association. The DeSurv high-risk group was significantly enriched for both Basal-like (PurIST) and proCAF (DeCAF) subtypes, consistent with the known poor prognosis of these molecular classes. By contrast, standard NMF risk groups at the matched rank ($k = 3$) separated Basal-like from Classical tumors (PurIST) but showed no significant enrichment for proCAF versus restCAF (DeCAF), indicating that survival-unsupervised factorization does not recover the tumor–stroma axis captured by DeSurv's D1. This concordance confirms that DeSurv's survival-driven factorization recovers clinically relevant biology without requiring subtype labels during training.

18 Standard NMF factor structure at independently selected rank

Throughout the main text, both DeSurv and standard NMF were fit at the same rank ($k = 3$), enabling a controlled comparison of the effect of survival supervision on factor structure. When each method selects its own rank independently, standard NMF chooses $k = 5$ via residual elbow detection or $k = 7$ via Bayesian optimization with $\alpha = 0$, while supervised DeSurv BO selects $k = 3$. We examine here the gene program structure at elbow-selected $k = 5$ for both DeSurv (Fig. S8) and standard NMF (Fig. S9), and the NMF $k = 7$ structure to illustrate additional fragmentation when supervision is absent (Fig. S10).

At elbow-selected $k = 5$, supervised and unsupervised factorizations yield substantially different gene-program structures. DeSurv at $k = 5$ (Fig. S8) splits the ECM/activated-stroma (proCAF-associated) signal into two factors with opposing tumor correlations (one positively correlated with Classical tumor programs, one negatively correlated) and isolates restCAF as a separate factor; exocrine and Basal-like programs are not represented as dominant factors in the supervised solution. Standard NMF at $k = 5$ (Fig. S9), by contrast, recovers the three core programs from its $k = 3$ solution and adds two factors capturing Exocrine-like and restCAF variance, transcriptional variance not aligned with survival outcomes. These biologically redundant NMF factors explain why supervised BO selects $k = 3$ in the production model: the additional NMF factors at $k = 5$ contribute no independent prognostic content beyond the three core programs (Fig. S11), even though the overall higher-rank model still validates externally on per-cohort C-index (Table S4).

At BO-selected $k = 7$ under $\alpha = 0$ (Fig. S10), standard NMF shows increased fragmentation relative to $k = 5$: multiple factors partially overlap the same PDAC gene signatures, and no new biologically coherent program emerges beyond those present at $k = 3$.

Standard NMF at elbow-selected $k = 5$ and BO-selected $k = 7$ both substantially improve over NMF at $k = 3$ across all cohorts (Table S4), confirming that additional unsupervised factors capture prognostically relevant variation missed at the matched rank. At $k = 7$, NMF matches or exceeds DeSurv's per-cohort C-index in most cohorts, demonstrating that unsupervised factorization can approach supervised concordance given enough factors. However, this comes at the cost of a less parsimonious and less interpretable factorization: the $k = 7$ NMF solution shows increased fragmentation (Fig. S10). DeSurv's advantage thus combines biological concordance and methodological complementarity: its risk groups identify the same poor-prognosis subtypes as established classifiers (Basal-like, proCAF; Fig. S7), while its linear predictor retains significant prognostic value after adjustment for those classifiers (main-text Table 1).

Table S5 provides a complementary perspective that isolates the prognostic information carried by each method's factor decomposition from the transfer quality of its trained linear predictor. Whereas main-text Table 1 freezes the training-derived β coefficients into a single risk-score covariate, Table S5 re-estimates β on each validation cohort using the individual factor scores (F1, F2, F3 for DeSurv; F1–F7 for NMF) as separate covariates and reports the Bayesian Information Criterion (BIC) of the resulting Cox proportional hazards fits. BIC uses $k \log n_{\text{events}}$ as the complexity penalty (Volinsky and Raftery, *Biometrics* 52, 203–223, 1996; Kass and Raftery, *JASA* 90, 773–795, 1995). DeSurv's parsimonious 3-factor decomposition is BIC-preferred over NMF's 7-factor decomposition in three of five validation cohorts (Dijk, Moffitt, PACA-AU array), with the remaining two cohorts essentially tied: NMF's four additional factors do not provide sufficient log-likelihood improvement to overcome their complexity penalty.

The reconstruction–survival decomposition at $k = 5$ (Fig. S11) confirms that elbow selection does not recover additional prognostic structure. At $k = 5$, standard NMF continues to distribute survival contribution negligibly across all factors; the two factors new relative to $k = 3$ (N4 and N5) contribute minimally, confirming that the additional rank captures background transcriptional variance rather than independent prognostic programs.

19 Factor-Specific Gene Lists

Tables S6–S8 list the top 270 genes for each of the three DeSurv factors, ranked by the factor specificity score s_{ij} defined in SI Appendix, Section 9. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ($n_{\text{top}} = 270$); these are the genes used to compute sample-level factor scores for survival prediction. Factor D1 is enriched for Classical tumor and restCAF stromal genes, Factor D2 for proCAF-associated stromal genes, and Factor D3 for Basal-like tumor and proliferation-associated genes.

20 Translational analysis in independent bulk treated PDAC cohorts (RASH-ACCEPT and Linehan)

The primary analysis evaluated DeSurv in treatment-naive resected primary PDAC, the setting in which the gene-program matrix W and Cox coefficients β were trained. As an exploratory translational test we asked whether the trained programs retain prognostic value in treated PDAC, and specifically whether they remain prognostic after adjustment for the established PurIST (tumor-intrinsic) and DeCAF (stromal) molecular classifiers — the same dual adjustment applied to the treatment-naive validation cohorts in the main paper (Table 1). We used three bulk-tissue treated cohorts spanning two disease settings: the two metastatic gemcitabine-backbone trials Rash ($n = 85$, gemcitabine + erlotinib) and Accept ($n = 70$, gemcitabine $\pm$ afatinib) — together the RASH-ACCEPT series ($n = 155$, bulk RNA-seq, first-line chemotherapy) — and Linehan ($n = 66$ borderline-resectable / locally advanced PDAC, bulk biopsy; FOLFIRINOX $\pm$ the CCR2 inhibitor PF-04136309). All are bulk tissue, so PurIST and DeCAF are both computable and the full dual adjustment is valid — unlike microdissected cohorts, where the stromal classifier cannot be applied. Because the cohorts differ in stage and regimen, they are analysed per cohort rather than pooled (below).

20.1 Per-cohort overall survival (no pooling)

Factor scores were computed by projection using the same trained, top-gene-truncated $\widetilde{W}$ ($n_{\text{top}} = 270$ per factor, union across factors) and within-sample rank convention as the main-paper validation cohorts ($Z_i = \widetilde{W}^\top X_{\text{new}}$), z-standardized within cohort; results were near-identical under a full-gene projection (Spearman $r \geq 0.86$; see Robustness below). Because the three treated cohorts differ in disease setting and regimen (Linehan borderline-resectable/locally-advanced on FOLFIRINOX$_\pm$CCR2 inhibitor; Rash and Accept metastatic on gemcitabine backbones), and the basal–classical program D_1 is heterogeneous across them, we report each cohort separately rather than as a pooled estimate. D_1 is reported *marginally*: because D_1 encodes the basal–classical contrast and is collinear with the PurIST classifier (it separates PurIST calls at AUC 0.84–0.89), adjusting D_1 for PurIST conditions the same axis on itself and over-adjusts; the marginal estimate is the appropriate measure of its prognostic value. D_2 (a stromal program, not collinear with the classifiers) is reported both marginal and PurIST + DeCAF-adjusted.

The proCAF stromal program D_2 was protective in all three cohorts (marginal and, where powered, adjusted), with a consistent direction across the independent settings and chemotherapy backbones; it was significant in Linehan and non-significant but directionally concordant in the metastatic cohorts (Accept and Rash). Because D_2 is stromal and not collinear with the classifiers, its association was not an artifact of adjustment: in Linehan it was significant both before and after adjustment, and across all three cohorts the adjusted estimate strengthened rather than attenuated relative to the marginal one, indicating the classifiers partially mask rather than create it. The basal–classical program D_1 was prognostic in the less-advanced Linehan cohort (marginal HR 0.58 (0.36–0.94)) but not in the metastatic cohorts — a setting-dependence we show below is a property of the basal–classical axis itself (shared by PurIST, Moffitt, and Bailey), not a failure of D_1. D_3 (Basal-like) was not associated with overall survival in any treated cohort, consistent with the basal–classical contrast residing in D_1.

20.2 The basal–classical axis is setting-dependent (classifier triangulation)

That D_1 attenuates in metastatic disease reflects the basal–classical prognostic axis itself rather than D_1 specifically: every established basal–classical classifier shows the same pattern (Table S10). In the less-advanced Linehan cohort the axis is strongly prognostic (PurIST Classical-vs-Basal HR 0.07 (0.02–0.26)), whereas in the metastatic cohorts PurIST, Moffitt, and Bailey are all non-significant. Basal-like prevalence in the metastatic cohorts is the expected 16–22% (not range-restricted; the Linehan estimate rests on only six Basal-like cases and is correspondingly imprecise), and the same axis is prognostic at $P < 10^{-8}$ in equally-prevalent treatment-naive validation cohorts (main text), indicating a genuine stage-dependent attenuation rather than reduced power.

20.3 D2 is prognostic on progression-free survival and is treatment-remodelled

In the metastatic RASH-ACCEPT cohort, D_2 was also associated with longer progression-free survival (HR per SD 0.80 (0.65–0.98), 0.031); D_3 was significantly associated with shorter progression-free survival (HR 1.35 (1.08–1.69), 0.009) — a single-cohort, single-endpoint observation consistent with the aggressive biology of Basal-like tumours. In the 29 Linehan patients with paired pre- and post-treatment biopsies, D_2 increased significantly from pre- to post-treatment (paired Wilcoxon 0.013; mean $\Delta = +0.34$ SD), indicating that the proCAF program is itself remodelled by therapy.

20.4 D2 molecular sub-components (exploratory)

Partitioning D_2's projection additively into gene-group contributions and testing each against overall survival (cohort-stratified; sub-scores are correlated, so the joint attribution is interpreted cautiously) localised the protective association mainly to the program's neural-reactive/axon-guidance content (Table S11): the neural-reactive component is protective both alone and in the joint model that controls for collinearity, whereas the extracellular-matrix-core component is protective alone but absorbed jointly, and the CD163$^+$ macrophage component is non-protective alone and, after joint adjustment, significantly adverse (HR 1.25, $P = 0.043$), consistent with reports that cancer-associated-fibroblast–driven macrophage reprogramming is tumour-promoting[28,29].

20.5 Orthogonal validation of program identity

Independently of survival, the D_1 program score correlated with GATA6 RNA in situ hybridization in the COMPASS cohort (Spearman $\rho = 0.68$, < 0.001, $n = 33$), validating the Classical/GATA6 identity of D_1 beyond the signature-level correlations in the main-text factor–signature heatmap.

20.6 Robustness to projection convention

The treated-cohort scores were computed with the top-gene $\widetilde{W}$ used in primary validation. Recomputing them with the full trained W (all 1970 genes) gives near-identical per-sample scores (pooled Spearman correlation D_1 0.86, D_2 0.98, D_3 0.96) and leaves every per-cohort conclusion unchanged, so the treated findings do not depend on the projection convention. We report results per cohort and do not pool: the protective direction of D_2 is consistent across all three cohorts, which is the relevant cross-cohort evidence in the absence of a common-effect assumption.

20.7 Caveats

This analysis is exploratory and prognostic, not predictive (treatment is confounded with cohort and stage, and biopsies are predominantly pre-treatment). (i) The cohorts are analysed per cohort and not pooled, because their settings differ (metastatic Rash/Accept vs borderline-resectable/locally-advanced Linehan) and D_1 is heterogeneous across them; the consistent protective direction of D_2 across all three cohorts is the cross-cohort evidence. (ii) D_1 is reported marginally because it is collinear with the PurIST classifier, so adjusting it for PurIST over-adjusts the same basal–classical axis. (iii) The treated cohorts are modestly sized (Linehan $n = 66$, 22 events; the metastatic Rash and Accept cohorts $n = 85$ and 70) so per-cohort D_2 estimates are individually under-powered, although the protective direction is consistent across all three. (iv) No analysis plan was preregistered. (v) The D_2 direction reversal in treated disease relative to treatment-naive primaries is the central context-dependent observation (Discussion); the sub-component decomposition and the established restraining-CAF literature provide candidate explanations, but mechanism is not established here. Robust treated-cohort validation will require larger, harmonized bulk-tissue series, ideally in the neoadjuvant setting where treated and resectable disease coincide.

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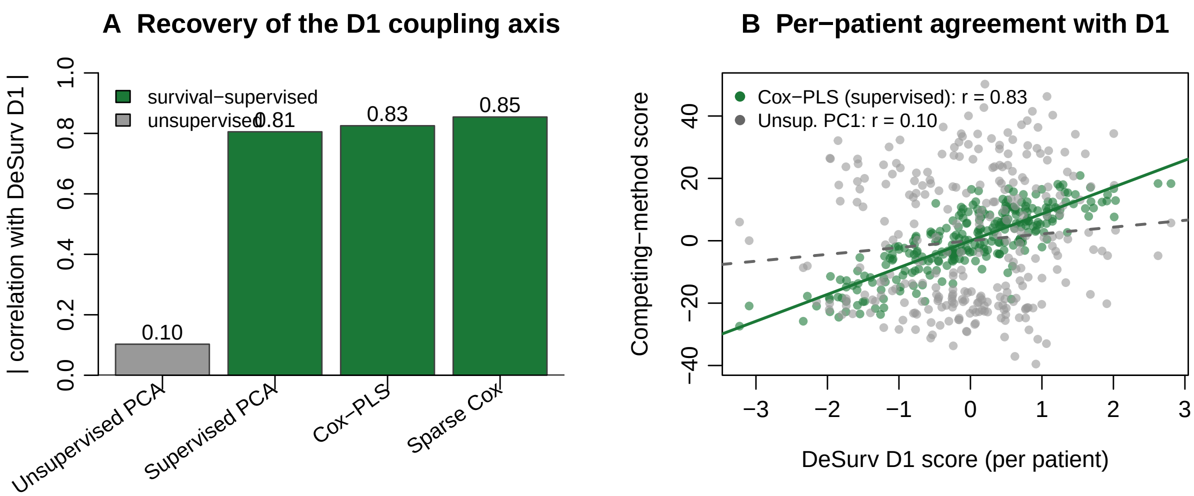

Figure S1: Independent survival-supervised methods recover the D1 axis; an unsupervised method does not. (A) Absolute per-patient correlation of each method's leading direction with the DeSurv D1 score, in the training cohort: supervised principal components, Cox partial least squares, and sparse Cox all recover D1 (green), whereas the leading unsupervised principal component does not (grey). (B) Per-patient agreement with D1 for Cox-PLS (supervised, green) versus the first unsupervised principal component (grey).

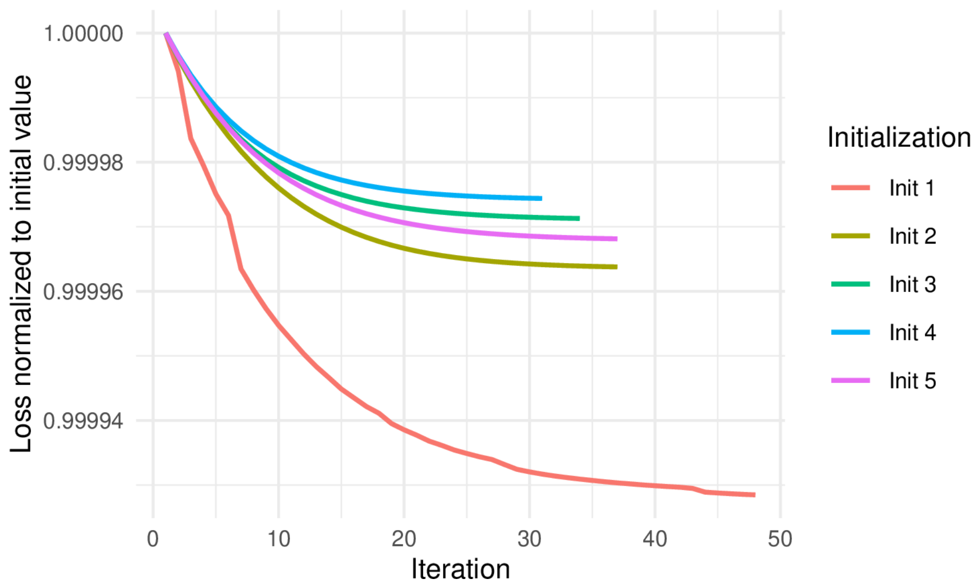

Figure S2: Convergence trajectories of the DeSurv objective across five randomly selected multi-start initializations on the PDAC training cohort (TCGA + CPTAC). Each line shows the relative loss (loss at iteration t divided by the initial loss for that restart) as a function of iteration number, restricted to the first 5,000 iterations for readability. The total objective decreases monotonically and stabilizes well within the iteration budget across all initializations, confirming convergence of the alternating block-update scheme.

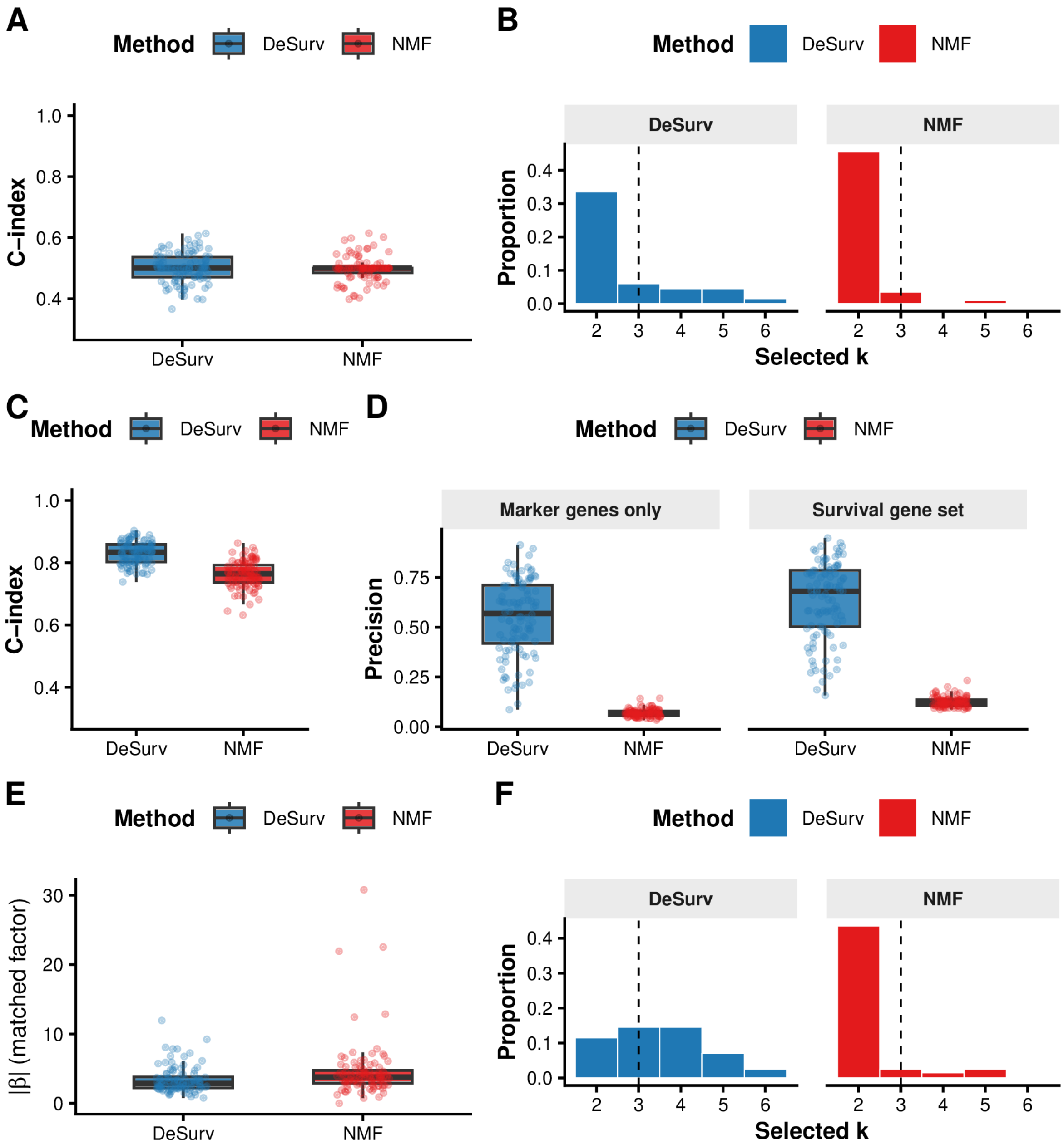

Figure S3: DeSurv performance under null and mixed simulation scenarios. (A–B) Null scenario ($\beta = 0$): both methods yield C-index ≈ 0.5 (A) and comparable rank distributions (B), confirming DeSurv does not hallucinate prognostic structure; precision is omitted as no true survival genes exist. (C–F) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (C) C-index is modestly higher for DeSurv. (D) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (E) The matched factor receives a substantially larger Cox coefficient $|\beta|$ under DeSurv, confirming preferential amplification of marker-specific signal. (F) Selected-rank distributions. DeSurv (blue); NMF (red, $\alpha = 0$); both Bayesian-optimized.

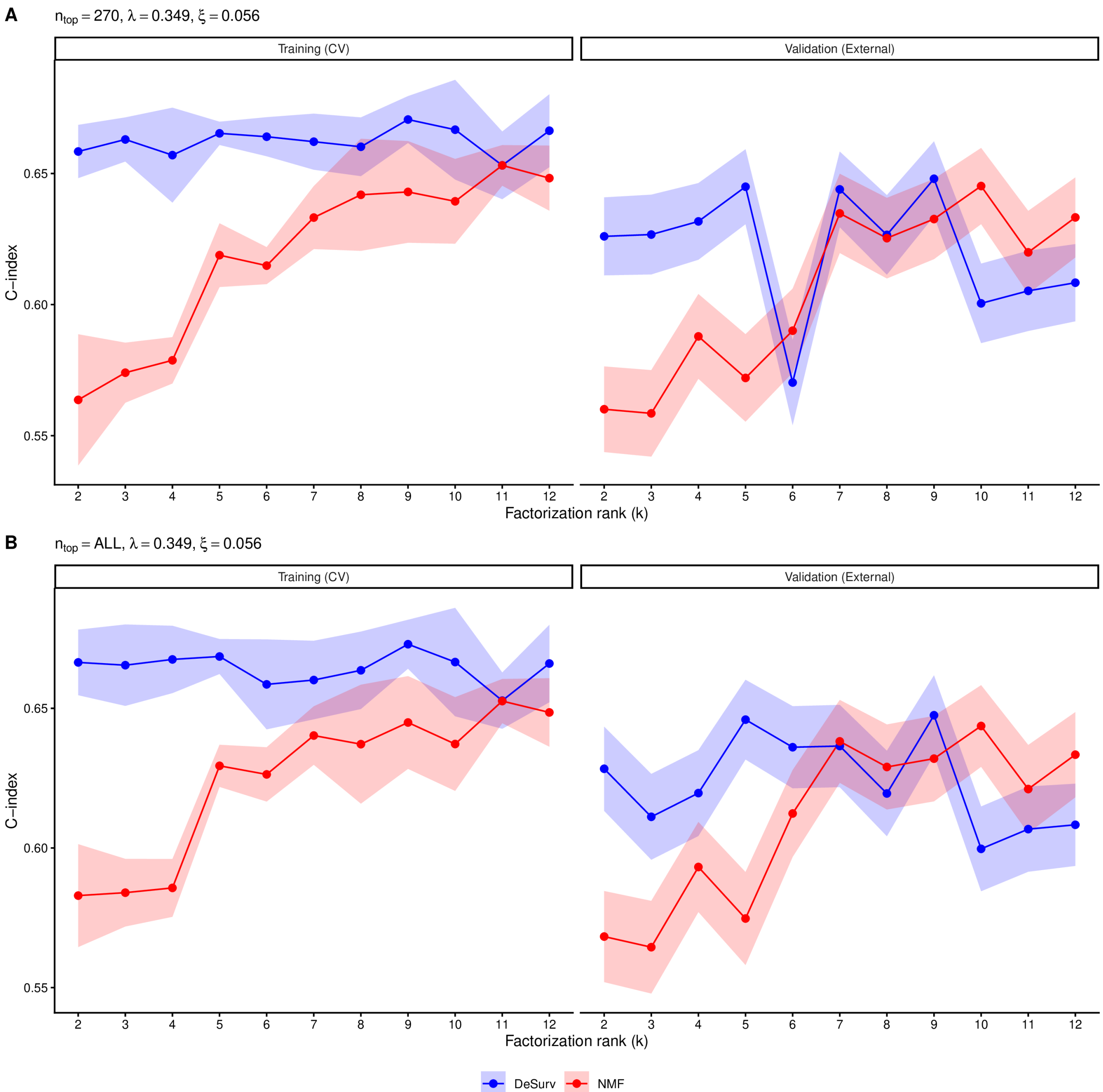

Figure S4: Training (CV) and external validation C-index as a function of factorization rank k for DeSurv (blue) and standard NMF (red, $\alpha = 0$). Shaded ribbons show ± 1 SE. For each k, the best α was selected by maximizing the mean CV C-index. Validation C-index is pooled across all external cohorts. (A) Results using $n_{\text{top}} = 270$ top genes per factor ($\lambda = 0.349$, $\xi = 0.056$). (B) Results retaining all genes ($n_{\text{top}} = \text{ALL}$; $\lambda = 0.349$, $\xi = 0.056$).

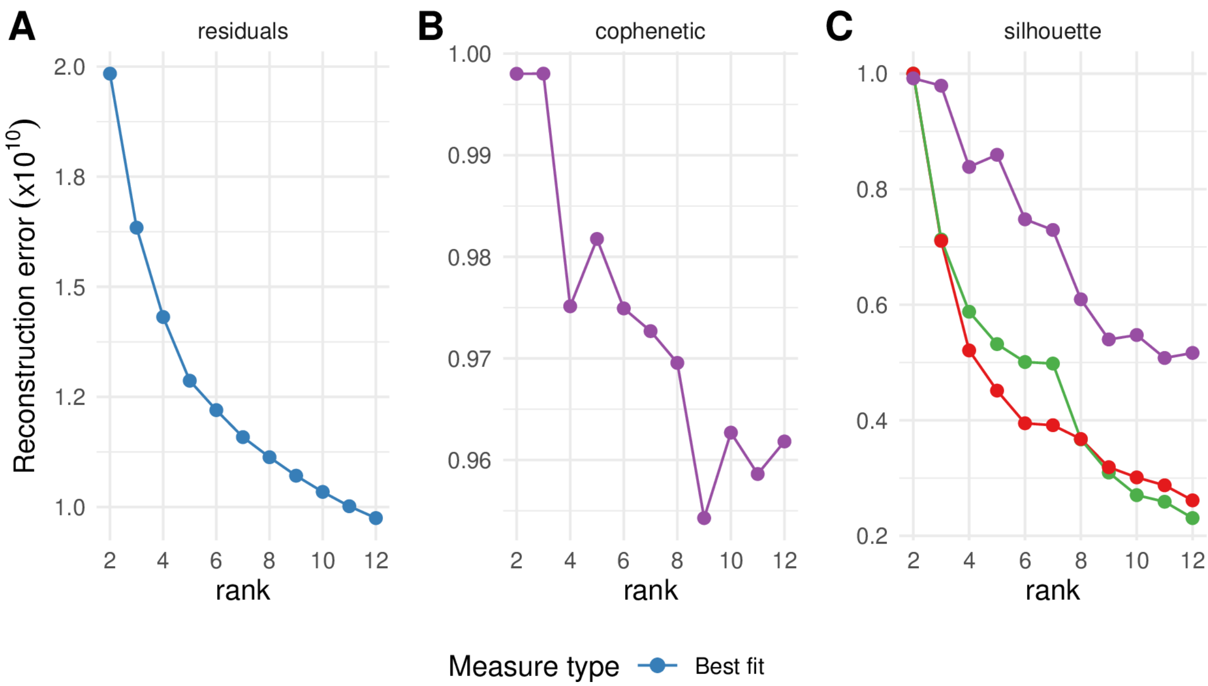

Figure S5: Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (A) Reconstruction residuals as a function of factorization rank k. (B) Cophenetic correlation coefficient as a function of k. (C) Mean silhouette width across multiple distance metrics as a function of k. These criteria yield inconsistent guidance, motivating the survival-supervised model selection approach (main text Fig. 5D).

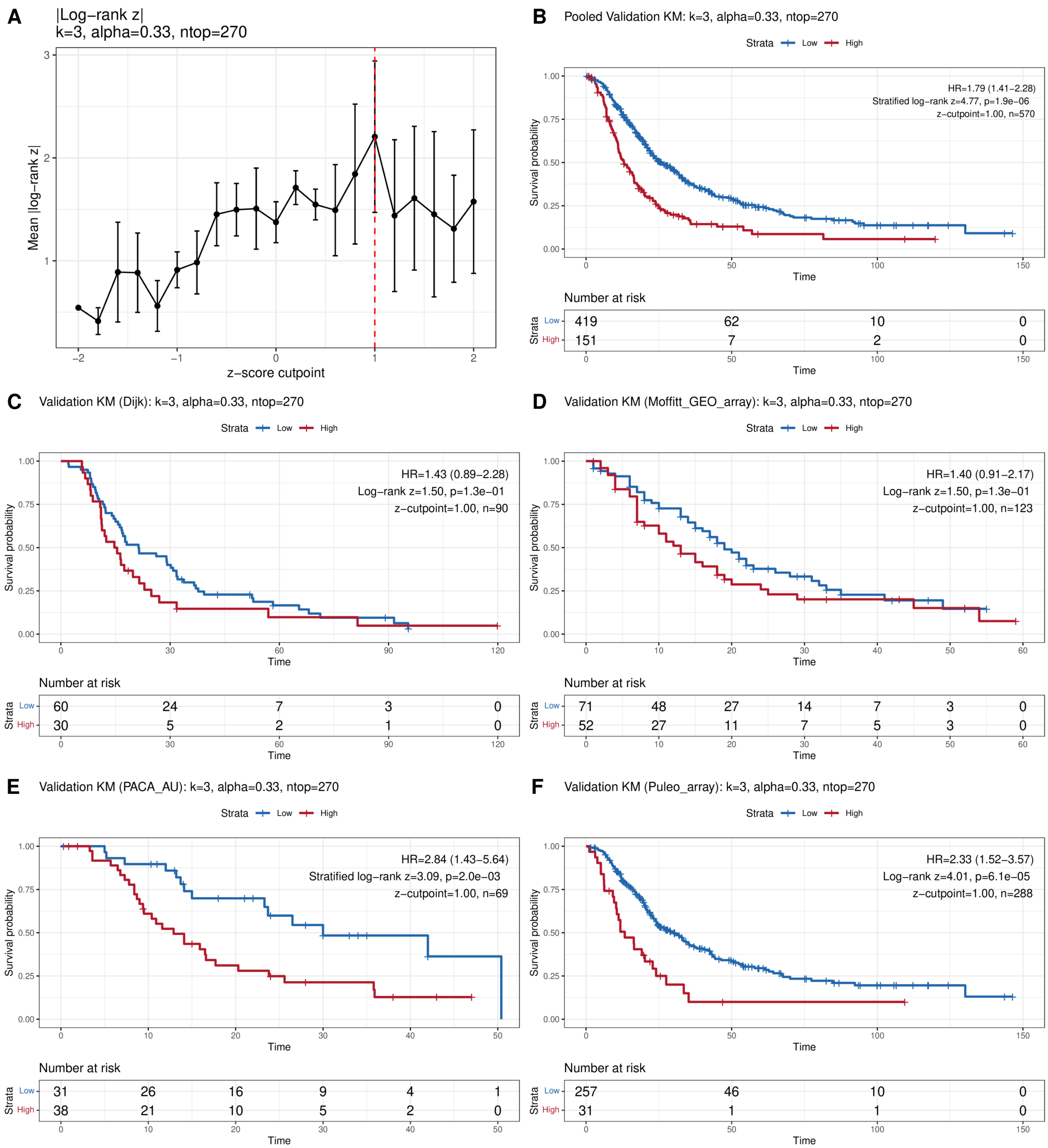

Figure S6: Cutpoint selection and external validation of dichotomized risk groups. (A) Cross-validated mean absolute log-rank z-statistic as a function of z-score cutpoint; red dashed line indicates the selected optimum. (B) Pooled validation Kaplan–Meier curves (stratified log-rank test). (C–D) Per-cohort validation KM curves: (C) Dijk, (D) Moffitt. (E–F) Per-cohort validation KM curves: (E) PACA-AU (array and RNA-seq subsets pooled), (F) Puleo.

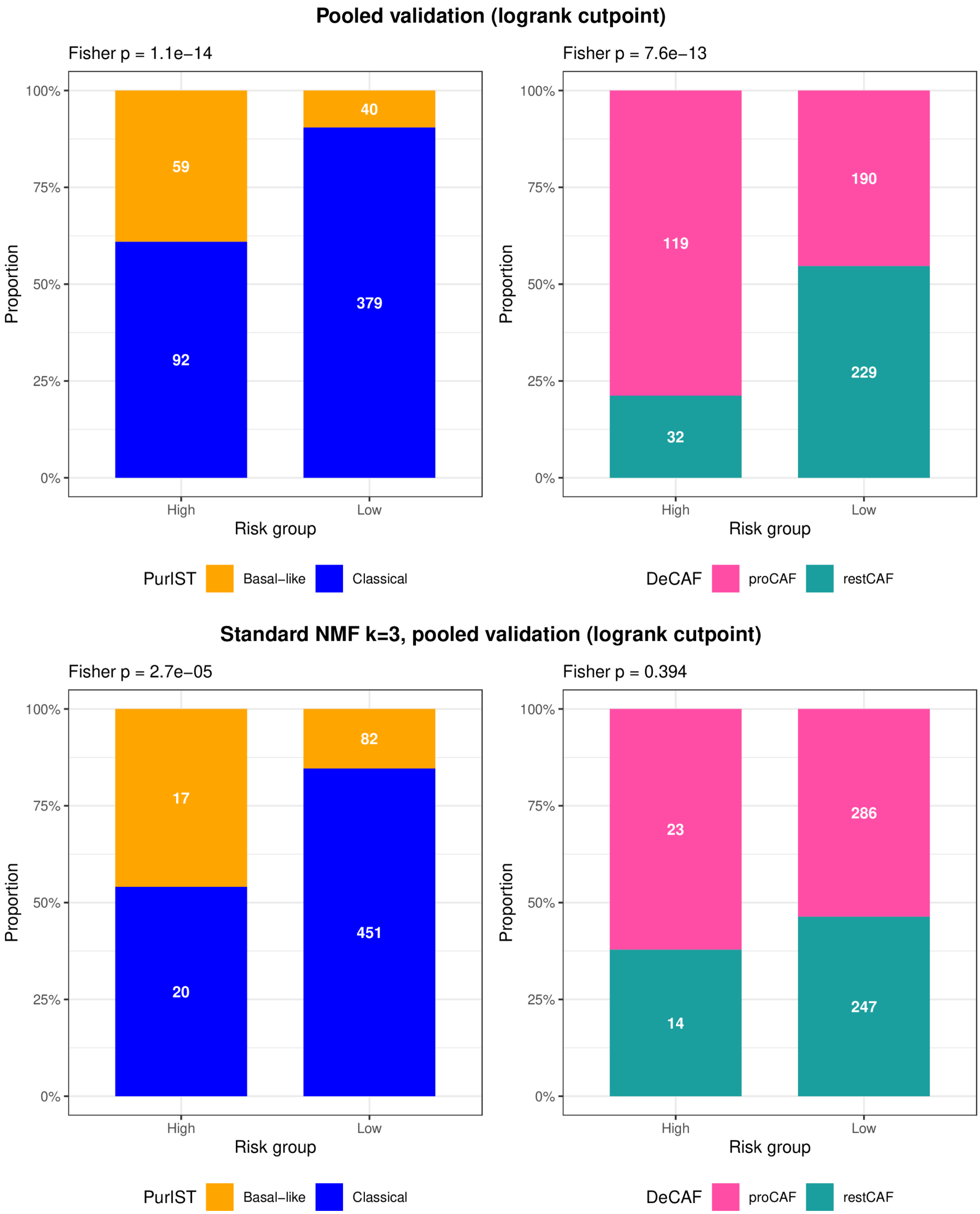

Figure S7: Subtype composition of DeSurv (top) and standard NMF $k = 3$ (bottom) risk groups in pooled external validation cohorts. Stacked bar charts show the proportion of PurIST (left) and DeCAF (right) subtypes within Low and High risk groups defined by the log-rank-optimized cutpoint. Numbers indicate sample counts; Fisher's exact test P-values are shown above each panel. DeSurv risk groups are enriched for both Basal-like (PurIST) and proCAF (DeCAF) poor-prognosis subtypes; NMF risk groups separate Basal-like from Classical tumors but show no significant CAF subtype enrichment.

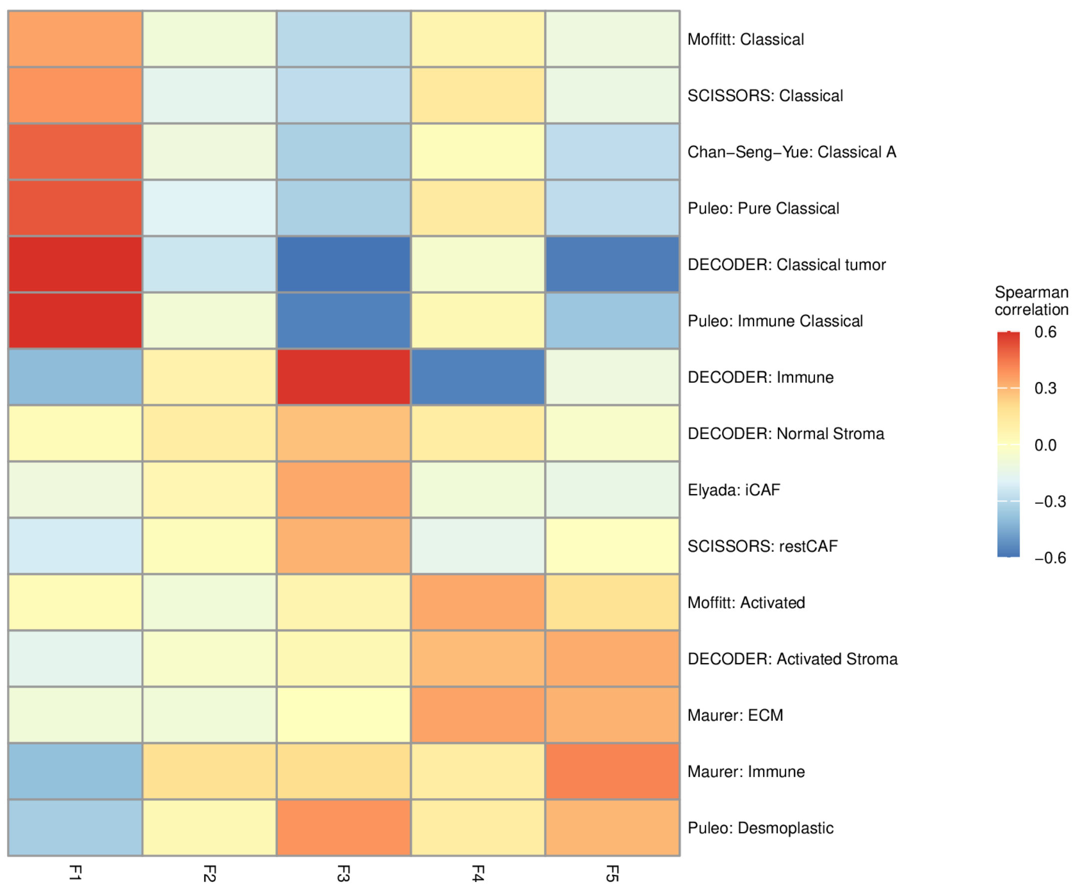

Figure S8: Gene program overlap heatmap for DeSurv at elbow-selected rank $k = 5$. Each cell shows the number of top genes from a given factor (rows) overlapping established PDAC gene programs (columns). DeSurv at $k = 5$ splits the ECM/activated-stroma (proCAF-associated) signal into two factors with opposing Classical-tumor correlations and isolates restCAF as a separate factor; exocrine and Basal-like programs are not represented as dominant factors.

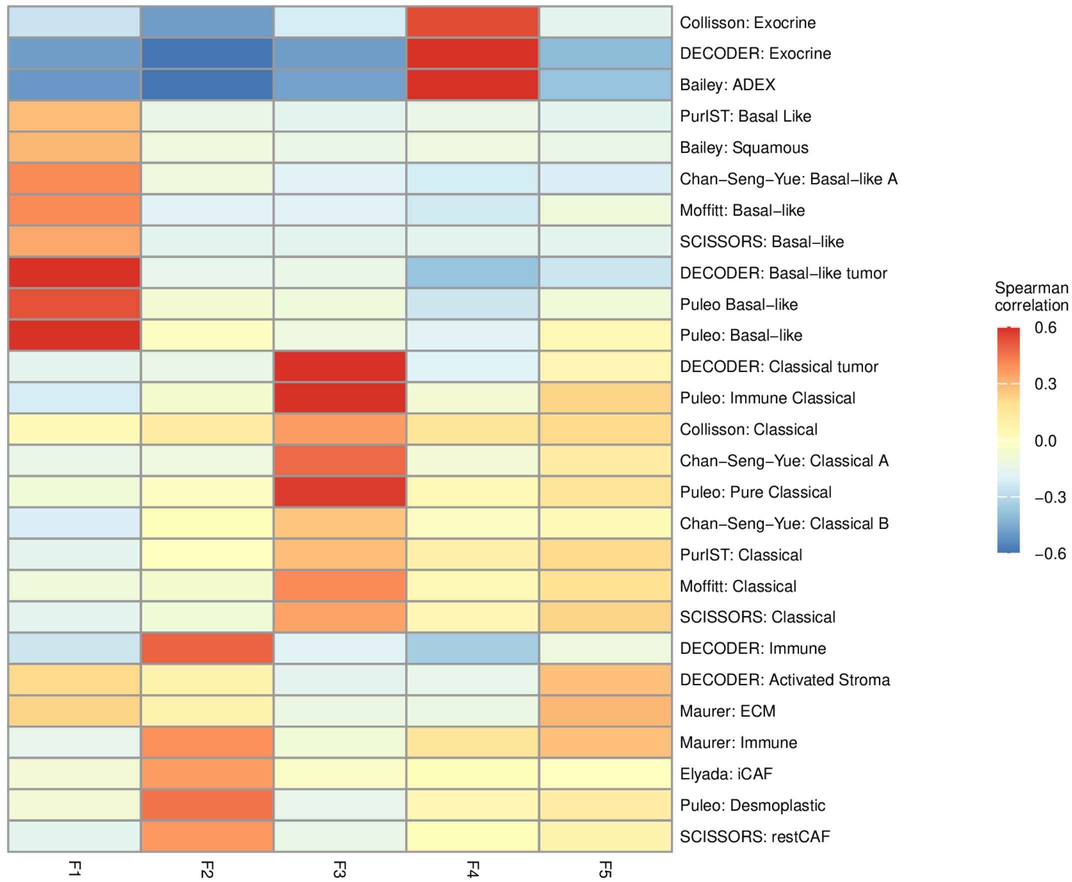

Figure S9: Gene program overlap heatmap for standard NMF at elbow-selected rank $k = 5$. Each cell shows the number of top genes from a given factor (rows) overlapping established PDAC gene programs (columns). The three core programs from $k = 3$ are recovered; two additional factors overlap Exocrine-like and restCAF signatures, capturing transcriptional variance not aligned with survival outcomes.

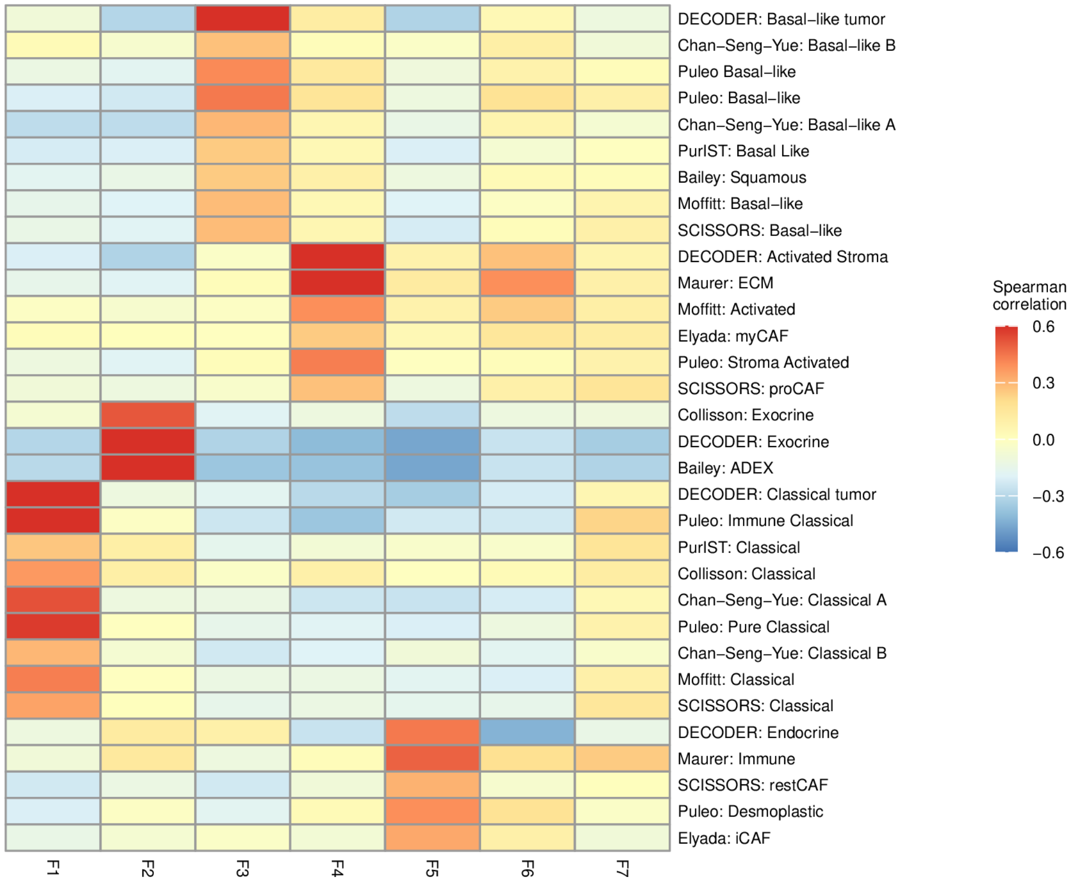

Figure S10: Gene program overlap heatmap for standard NMF at $k = 7$ (BO-selected rank with $\alpha = 0$). Compared with $k = 3$ and $k = 5$, the 7-factor solution shows increased fragmentation: multiple factors partially capture the same PDAC gene signatures and no new coherent biological program emerges.

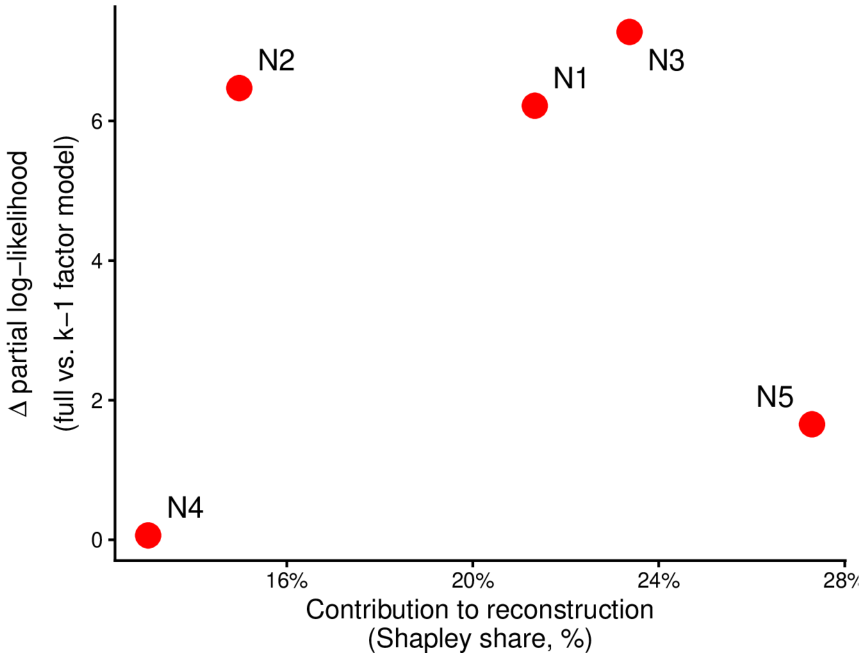

Figure S11: Per-factor contribution to reconstruction (normalized reconstruction Shapley share) versus Type III partial likelihood contribution for each standard NMF factor at elbow-selected $k = 5$. Each point represents one factor, labeled N1–N5. Survival contribution is quantified as the reduction in Cox partial log-likelihood when each factor is removed from the full model. The two factors new relative to $k = 3$ (N4, N5) contribute negligibly to survival, confirming that elbow selection recovers no additional prognostic structure.

Table S1: Summary of PDAC cohorts used in this study. N denotes the number of samples after all quality-control filters (histology, survival validity, dataset-specific exclusions). Patient-level metadata: OS = overall survival (time and event indicator); PurIST = tumor subtype classification (Rashid et al. 2020); DeCAF = CAF subtype classification (Peng et al. 2020). Expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study (Peng et al. 2020).

Cohort	Platform	N	Role	Accession	Metadata	Reference
TCGA-PAAD	RNA-seq	144	Training	GDC (TCGA-PAAD)	OS, PurIST, DeCAF	Raphael et al. (2017)
CPTAC	RNA-seq	129	Training	GDC (CPTAC-3)	OS, PurIST, DeCAF	Ellis et al. (2013)
Dijk	RNA-seq	90	Validation	E-MTAB-6830	OS, PurIST, DeCAF	Dijk et al. (2020)
Moffitt	Microarray	123	Validation	GSE71729	OS, PurIST, DeCAF	Moffitt et al. (2015)
PACA-AU (array)	Microarray	63	Validation	GSE36924	OS, PurIST, DeCAF	Bailey et al. (2016)
PACA-AU (RNA-seq)	RNA-seq	52	Validation	EGAS00001000154	OS, PurIST, DeCAF	Bailey et al. (2016)
Puleo	Microarray	288	Validation	E-MTAB-6134	OS, PurIST, DeCAF	Puleo et al. (2018)
Training total		273				
Validation total		616				

Table S2: Cross-method projection R^2: how reconstructable each factor's gene-loading vector is from the other method's full basis (OLS R^2). DeSurv retains the bulk of the unsupervised structure (NMF N1 and N3 carry over at $R^2 \geq 0.88$) and reorganizes it: the survival-aligned D1 program is essentially unreconstructable from the NMF basis ($R^2 = 0.09$), i.e. created de novo by supervision, while the exocrine NMF factor N2 dissolves ($R^2 = 0.38$).

Factor	R^2 from other basis	Interpretation
NMF N1 (tumor)	0.90	preserved
NMF N2 (exocrine)	0.38	dissolved
NMF N3 (microenviron.)	0.88	preserved
DeSurv D1 (Classical/restCAF)	0.09	novel (≈91% de novo)
DeSurv D2 (proCAF)	0.81	inherited (≈N3)
DeSurv D3 (Basal-like)	0.75	inherited (≈N1)

Table S3: Whole-matrix reconstruction R^2 for unsupervised NMF versus DeSurv at $k = 3$. DeSurv sacrifices only ≈ 1.5 percentage points of reconstruction R^2 to gain its prognostic structure.

Model	Reconstruction R^2	Residual fraction
NMF ($k = 3$)	0.953	0.047
DeSurv ($k = 3$)	0.938	0.062

Table S4: External validation C-index by method and cohort. DeSurv at BO-selected rank ($k = 3$) is compared with standard NMF at matched rank ($k = 3$), elbow-selected rank ($k = 5$), and BO-selected rank ($k = 7$, $\alpha = 0$). Higher values indicate better discrimination.

Cohort	DeSurv (K=3)	Std NMF (K=3)	Std NMF (K=5, elbow)	Std NMF (K=7, BO α=0)
Dijk	0.607	0.542	0.633	0.631
Moffitt_GEO_array	0.562	0.521	0.552	0.578
PACA_AU_array	0.649	0.470	0.657	0.669
PACA_AU_seq	0.634	0.430	0.679	0.702
Puleo_array	0.636	0.534	0.642	0.651

Table S5: Per-cohort Bayesian Information Criterion (BIC) for Cox proportional hazards models using individual factor scores as predictors of overall survival in each external validation cohort. β coefficients are re-fit on each validation cohort separately. DeSurv uses three factor scores (F1, F2, F3); NMF at BO-selected $k = 7$ uses seven (F1–F7). BIC penalises complexity as $k \log n_{\text{events}}$. Positive ΔBIC (NMF − DeSurv) indicates DeSurv's parsimonious 3-factor decomposition is preferred. DeSurv is BIC-preferred in three of five validation cohorts; the remaining two are essentially tied.

Cohort	n	events	DeSurv ($k = 3$) BIC	NMF ($k = 7$) BIC	Δ BIC
Dijk	90	81	598.4	605.8	+7.4
Moffitt GEO array	123	83	674.8	686.0	+11.2
PACA-AU array	63	38	265.6	278.5	+12.9
PACA-AU RNA-seq	52	31	197.8	199.3	+1.5
Puleo array	288	181	1782.0	1782.6	+0.6

Table S6: Top 270 genes for DeSurv factor D1 (Factor D1 (Classical tumor + restCAF stroma)), ranked by factor specificity score s_{ij}. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension (n_{top}).

Factor D1 (Classical tumor + restCAF stroma)					
1–45	46–90	91–135	136–180	181–225	226–270
DMBT1	FGFR2	GOLGA8A	PTPRN2	RPL9	PPP1R12B
IGFBP2	COL9A2	IL2RG	FCGBP	CD79A	PIP5K1B
S100A4	COL22A1	KIAA1211	PLA2G2A	PROM1	AHCYL2
KRT23	IRF7	SFRP1	FABP4	SCD	SST
LIF	SBNO2	ADH1B	EPPK1	TMC5	CYP3A5
IFI6	MGAT3	MFSD2A	CPXM2	CCL14	SMAP2
PDLIM3	SYNM	CLU	KIF1A	PLXNB1	ZG16B
CLDN18	APOD	CLMN	NGFR	TMEM119	SERPINF1
MYO15B	SEMA6A	CHPF	NFKBIA	SORBS2	CLDN3
SYT8	TNXB	MTMR11	SPOCK2	C7	NR1I2
TFF2	PTGIS	TTR	ATP2C2	SLC12A7	JUND
TMPRSS2	CCL18	WFDC2	NBEAL2	QSOX1	NCF2
CHGB	ZC3H12A	C2orf72	CAPN9	UNC5CL	CELSR1
CST1	CELSR3	CCND2	PTGDS	LONRF2	MUC17
B3GNT7	EMB	ARHGAP4	LSP1	VSIG2	INPP5D
F5	KCNJ15	PKDCC	LAMA5	ADAMTSL2	TM4SF5
TNRC18	SORBS1	SRPX	FASN	ARHGEF16	HNF4A
IGF2	PAQR8	CCL21	GFRA1	PCOLCE	CFD
ARSD	SLC4A4	CARD11	PTPRH	CFC1B	HCLS1
ZDHHC7	ACSS1	CTRB2	FMOD	SLC9A3	SH3PXD2B
LENG8	ST6GALNAC1	GOLGA8B	FHL1	ACACB	PCSK2
ATP9A	DES	IYD	PCDH20	BCAS1	HMGCS2
TMPRSS3	PRUNE2	CORO1A	CCR7	RASD1	FILIP1L
CES1	PLCB4	SAA2	CHIT1	LGALS4	ETNK1
HOPX	CADPS	CDCA7	PARM1	MS4A1	TNNI2
CNN1	FMO5	BOC	ITM2A	CAPN6	TOX3
MIA	RAP1GAP2	LTB	NAPSB	SCCPDH	FRZB
CRIP2	GALNT12	COMP	AGT	PTPRCAP	HABP2
NEURL1B	CCL2	MANSC1	FUT3	VNN1	GDA
CHGA	REG3A	SLC19A3	SIPA1L2	PI16	NCOR2
ACTG2	ZCCHC24	STEAP2	SPINK5	SEMA7A	DNAH2
CXCL14	LTF	FOXA1	ACSL1	IL10RA	DOK4
TPPP3	EFHD2	WNK2	BTNL8	GALNT3	HOXB9
PIGR	ATP11A	SELL	CORO2A	GOLM1	KIAA1324
CYP27A1	GP2	PLEKHA6	LAD1	TNFAIP2	TRIM2
ADAM8	HIPK2	NFASC	TMEM63A	CD79B	FGFBP1
FGFR3	SLC26A9	EGR3	GC	ENPP2	IRX3
IGF1R	TACC2	PLIN4	C1orf21	SQLE	TINAGL1
PLEKHB1	TRAK1	WDR72	MIF	KLF4	SFRP4
CPXM1	SAA1	AMY2A	FOXA2	PDLIM7	APLNR
ATP7B	CLDN23	IL16	CCL19	FOSB	AGRN
SLC12A2	FOXJ1	TJP3	GSTA1	RHPN2	SLC25A25
CCL20	FRY	SLC9A2	LLGL2	SCG2	PODN
AKNA	GATA6	LMOD1	ACP5	EMILIN1	ITIH2
COL16A1	PLXNA2	CPS1	UGT2B15	BACE2	FLJ23867

Table S7: Top 270 genes for DeSurv factor D2 (Factor D2 (proCAF stroma)), ranked by factor specificity score s_{ij}.

Factor D2 (proCAF stroma)					
1–45	46–90	91–135	136–180	181–225	226–270
UHMK1	RPS26	SAMD9L	ATP8A1	IRF6	RASSF2
DDR2	MYLK	DOCK2	TGFBR1	KLHL6	ZNF83
INHBA	COL11A1	IGF1	SYNPO2	C1QTNF3	CYTIP
DYNC2H1	NFIB	TACC1	PDE4B	PCLO	SRPX2
ZDHHC20	PTPN13	CFH	SGCD	ITGA2	LY75
ITPR2	PDE3A	DMD	STK17B	LRIG3	HLA-DOA
HMCN1	TRIM22	PCDH7	NUAK1	SSPN	STK38L
LAMA2	DPYD	MALAT1	FCGR3A	F2R	RSAD2
SLFN5	COL14A1	NOTCH2	GPNMB	PDGFC	HBB
PLXNC1	CNTN1	CDK6	VCAM1	CD109	HERC2P2
SVEP1	NCKAP1L	FBN1	ADAMTS9	CLIC4	AOX1
CYBB	PHLDB2	SYNE2	ITGB8	SORL1	PLIN2
SYNE1	ITGA4	RHOBTB3	CR1	AOAH	PFKFB2
SAMHD1	MACC1	CALCRL	SEPT11	FMO2	OGN
SKIL	SEMA5A	NRP1	IFI44L	RDX	DOCK10
ADAMTS12	MRC1	LIFR	ELL2	IL1RAP	RGS4
BICC1	SLIT2	STAB1	GPR183	GBP1	AXL
ITGA1	ZEB2	FLRT2	ZBTB16	CXCR4	FGD6
COL8A1	ADAM10	RNF144A	FRMD6	WNK1	CDC42BPA
PDGFRA	SAMD9	BCAT1	PDE5A	MFAP5	SLC6A6
ABCA1	MS4A7	CILP	CSF1R	ALDH1A3	RGS1
TCF4	EFEMP1	CCDC80	PLEK	CD53	KIAA1324L
CEP170	BIRC3	FAT4	SLK	OMD	CPA3
CD163	WISP1	BNC2	SLC7A2	PLOD2	CHL1
MAN1A1	LAMA4	LTBP1	HEG1	ITGAV	SLC7A11
MSR1	FAM198B	COL12A1	GREM1	MACF1	GFPT2
CCDC88A	PTPRC	IL1R1	VGLL3	NID2	RAI14
F13A1	FAP	SLA	GBP5	NAMPT	LMAN1
RPS28	ARRDC3	MBOAT2	SLC1A3	ADAM28	MECOM
RALGAPA2	MYH10	FNDC3B	FGF7	GNE	FPR1
ZEB1	FCGR2A	PRRX1	ABCA8	ENAH	EDEM3
FBXO32	PARP14	COL10A1	ADAM12	FAM129A	RASSF6
INPP4B	DSE	DDX60	CYBRD1	ECM2	IQGAP2
CDH11	EHF	DOCK9	TFRC	VSIG4	MIAT
PKHD1	OSMR	SRGN	CTSK	MTAP	PRICKLE1
AKAP2	ARID5B	FOXO1	COLEC12	INMT	CHST15
CYP1B1	ASPN	MS4A6A	ALDH1L2	SGMS2	GEM
CCL5	SLCO2B1	ADAMTS2	PGM2L1	COL4A4	CHRDL1
FGL2	FPR3	CPVL	FAM135A	CAPRIN2	DSC2
MAP1B	DOCK8	CD93	DOCK11	CENPF	RAB31
PLXDC2	SLFN11	DST	EPHA3	DCLK1	SLC38A1
LRRK2	ZNF532	CDK14	PAPPA	EDNRA	CNN3
KIAA0754	SPON1	MSRB3	EGFR	WNT5A	FLT1
EDIL3	SEMA3C	RNF213	RASA1	GALNT5	TMEM200A
ITGBL1	FERMT2	SEL1L	AHR	SLC38A2	SLC2A3

Table S8: Top 270 genes for DeSurv factor D3 (Factor D3 (Basal-like tumor)), ranked by factor specificity score s_{ij}.

Factor D3 (Basal-like tumor)					
1–45	46–90	91–135	136–180	181–225	226–270
HSPB1	HLA-H	LAMB3	MYL9	IL1RN	SDC1
KRT7	HSPA1A	FSCN1	COL17A1	TST	KLK11
MSLN	LGALS3	TSPAN13	EPS8L1	RAPGEFL1	PERP
RPSAP58	FAM83H	TSPAN8	CDH3	C15orf48	ARHGAP23
S100P	EPB41L1	ALDH3B1	RNASET2	PFKP	HPGD
RPS16	EFNA1	ANPEP	TUBA1C	STARD10	AKR1C3
IL32	TKT	GSTP1	RAP1GAP	FOXQ1	CRABP2
SH3BGRL3	SLC2A1	KLK10	H19	REG4	GRB7
IFITM2	PLOD1	B4GALT1	LTBP3	GJB1	SERPINA4
RPS21	IER3	GAS6	AQP3	SLC29A1	SLC39A4
SEMA4B	B3GNT3	ABCC3	MPZL2	NOTCH3	FA2H
APOL1	PRSS3	CLDN2	ID1	TRIM16	RPL8
S100A16	TOB1	CEACAM6	RARRES2	TAP2	MT1X
DNAJB1	EGR1	CREB3L1	SLC25A6	KRT19	TCN1
MT2A	PLA2G16	MST1R	RBP1	CRP	AKAP1
TNFRSF21	OCIAD2	CEBPB	HSPA5	SNHG5	GJB3
RPL28	LPCAT4	TXN	PER1	KRT18	DUOXA2
FXYD3	BST2	NR4A1	ADAP1	CA12	HLA-DRB5
SERPING1	EPHA2	AP1M2	TMC4	MYEOV	ARRDC2
HMGA1	MUC5B	SYNPO	HSD17B2	P4HB	OAS1
S100A14	KCNN4	ANO1	PTP4A1	VILL	ADRA2A
H1F0	ITGA3	SIK1	PLAUR	HKDC1	NQO1
KRT17	TSTA3	EFNB2	ANXA10	STEAP3	LRG1
CSTB	IFI27	CDKN1A	PYGB	PLEK2	SERPINH1
ASS1	MMP1	S100A10	CYP2S1	CD68	COL18A1
EFNB1	SLPI	PKP3	ALDOA	MAFF	CERCAM
MDK	CLTB	MCM7	CDCP1	CD9	CA9
LY6E	SPINK1	RARRES3	IFITM3	PLLP	CDHR2
CAMK2N1	JUP	CEACAM1	BAIAP2L1	EPHB4	ESRP1
MGLL	PPP1R15A	ALDH2	MGST1	KRT16	CRYZ
PDZK1IP1	TRIM29	CES2	TFF1	ABLIM3	TAGLN2
BCL2L1	PPP1R1B	SLC6A8	MUC4	PTPRU	HNF1B
C19orf33	ARPC1A	PPL	MMP28	MUC6	BMP4
SFN	GPRC5A	NDRG1	CLDN1	FDFT1	MT1E
ITGB4	RAB25	AZGP1	TM4SF1	GAPDH	ANGPTL4
TSPAN1	ZFP36L2	S100A11	TFF3	EPS8L3	CFB
SLC16A3	CLDN7	SEMA3B	UNC13D	CLDN10	GDF15
C4A	DDIT4	TNFRSF12A	SERINC2	ITPKC	PADI1
KLF5	FKBP4	SOX9	SPNS2	LOXL2	TMSB10
IER2	LAPTM4B	PMEPA1	AQP5	AMIGO2	B2M
ZFP36L1	MUC13	CYR61	IGFBP4	DUOX2	CAV2
MAL2	PSCA	EPCAM	SLC7A5	CLDN4	CTSH
SCNN1A	SDCBP2	BHLHE40	TESC	CAPG	PABPC1
MALL	PROM2	CX3CL1	IFITM1	CEACAM5	KRT8
BGN	LRRC8A	SMAD3	KCNK5	VWA1	TAP1

Table S9: Per-cohort overall-survival hazard ratios per SD in the three bulk treated PDAC cohorts. D_1 marginal; D_2 marginal and PurIST + DeCAF-adjusted; D_3 marginal. Sample sizes: Linehan $n = 66$ (22 events), Rash $n = 85$ (73 events), Accept $n = 70$ (57 events); Basal-like prevalence Linehan 9% (6 cases), Rash 22%, Accept 16%.

Cohort (setting)	D_1 (marginal)	D_2 (marginal)	D_2 (+PurIST+DeCAF)	D_3 (marginal)
Linehan (BR/LA)	0.58 (0.36–0.94), 0.027	0.60 (0.42–0.86), 0.005	0.46 (0.31–0.70), < 0.001	1.09 (0.68–1.77), 0.715
Rash (metastatic)	0.81 (0.63–1.04), 0.096	0.87 (0.66–1.13), 0.283	0.77 (0.53–1.12), 0.178	1.09 (0.87–1.38), 0.455
Accept (metastatic)	1.10 (0.87–1.38), 0.427	0.78 (0.58–1.04), 0.091	0.73 (0.52–1.04), 0.078	1.11 (0.85–1.45), 0.445

Table S10: Classical-vs-Basal overall-survival hazard ratios for established subtype classifiers, by cohort. All are strongly prognostic in less-advanced disease (Linehan) and attenuate in metastatic disease (Rash, Accept).

Classifier	Linehan (BR/LA)	Rash (metastatic)	Accept (metastatic)
PurIST	0.07 (0.02–0.26), < 0.001	1.24 (0.72–2.16), 0.439	0.83 (0.41–1.71), 0.622
Moffitt	—	0.84 (0.51–1.39), 0.502	1.00 (0.49–2.05), 0.999
Bailey (squamous)	—	0.79 (0.46–1.34), 0.379	1.31 (0.70–2.45), 0.402

Table S11: Exploratory additive decomposition of the D_2 projection into gene-group sub-scores; per-SD overall-survival HR (cohort-stratified, PurIST + DeCAF-adjusted), each sub-score alone and all jointly.

D_2 sub-component	Alone HR/SD (95% CI), P	Joint HR/SD (95% CI), P
Neural	0.72 (0.59–0.88), 0.002	0.65 (0.49–0.87), 0.004
ECMcore	0.78 (0.63–0.96), 0.017	0.97 (0.75–1.26), 0.834
Macrophage	1.04 (0.86–1.24), 0.700	1.25 (1.01–1.54), 0.043